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A Data-Driven Study of Disney Films in Film Industry

Research Questions and Results

- How have the genres of Disney movies evolved over the years, and do these changes align with the overall trends in the global film and television industry?

answer: The types of Disney’s productions have gradually diversified over time and expanded rapidly after 2000. These changes generally align with the long-term trends of the global film industry, but Disney still maintains a unique brand preference centered on “family + fantasy,” which differs from the dominant comedy-oriented genre structure in the global market.

- How have audience perception and popularity of Disney movies evolved over time, and how do these trends compare with global movies when incorporating both numerical indicators (IMDb rating, votes) and linguistic features?

answer: Disney’s audience engagement has increased in line with global trends, but its rating fluctuations and language-emotional characteristics differ significantly from those of global films. This suggests that Disney is more strongly influenced by its internal creative strategies and brand narrative style, whereas global trends reflect a more long-term and stable structure of audience evaluation.

- Are there any relationships between runtime, genre, IMDb rating, awards, and Metascore? Does Disney movies show the same pattern as Global movies? (The analysis will not only explore whether these relationships are positive/negative, but also using statistical testing to determine the hypothesis.)

answer: Disney and global films exhibit significant differences in their rating systems: Disney’s audience ratings are more consistently influenced by Metascore, genre, and awards, whereas the rating patterns of global films are more diverse and complex. This indicates that Disney’s evaluation structure is more dependent on its brand characteristics, while the global market reflects more complex cultural and genre differences.

Motivation

With the rapid development of society, people’s lives have become increasingly rich and diverse, offering more ways to spend their time. Among these, film and television works have become a “necessity” in everyday life. Beyond individual hobbies, people are paying more and more attention to movies, TV series, and variety shows during their leisure time away from work and study. Therefore, in order to better understand and meet people’s entertainment needs, it is necessary to study the development trends of film and television works and the key characteristics of popular productions in recent years.

Furthermore, as a major source of economic growth, analyzing the trends and audience preferences for different genres can also contribute to promoting the development of the entertainment economy. The reason I chose to focus specifically on Disney movies is that Disney holds a significant influence within the global entertainment industry. As a Disney fan myself, I also have a personal curiosity to explore this topic more deeply. I have noticed that Disney movies have remained well-received over time, and I want to understand which themes are most popular, when they started gaining global attention, and how these patterns relate to trends in the broader film and television industry.

Moreover, most current research in society tends to focus on scientific or academic topics, while relatively few studies take a comprehensive look at people’s entertainment activities from both cultural and professional perspectives. Therefore, I believe this research is particularly meaningful. The findings may even provide insights that help guide adjustments in the film and television industry, ultimately contributing to a higher level of public happiness and cultural engagement.

Data Setting

Main dataset: Disney movies&shows <https://www.kaggle.com/datasets/eshummalik/disney> The dataset collects information about Disney shows (show names, types, release dates, ratings, directors, genre, awards, etc.), which was collected by a user/collector called Eshum_malik on Kaggle. According to the dataset, the three contexts that may potentially influence my analysis are: the missing data in early years (the shows lack complete information in earlier years, so this may cause a different result); the judgement of genre changes overtime (which means is rather subjective and not uniform); confidential business compact (rates or scores may not be purely rated by audience).

Additional dataset: IMDB movies (top 1000) <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows> For this dataset, it collects information about movies and tv shows that are within the top 1000. The content contains are similar to the Disney dataset: show names, release dates, ratings, directors, genre,votes, etc.), which was collected by a user/collector called Harshit Shankhdhar on Kaggle. According to the dataset, the three contexts that may

potentially influence my analysis are: the given condition (top 1000) data, which cannot fully represent movies all over the world; the time period covers a wide range, which have lots of influential factors to comparison, such as techniques; confidential business compact (rates or scores may not be purely rated by audience), similar as Disney dataset.

Method

Libraries that may be used: pandas, numpy, matplotlib, seaborn

Overview of steps:

1. Identify columns that the content inside needs to be revised in a uniform format
2. Create new columns, lists, or dictionaries in the dataset to facilitate data manipulation and plotting
3. Writing codes to get plots and models to study relationships between variables
4. Statistical tests & analysis

data-manipulation

- create a small dataset as testing_csv being used to test data manipulation.
- revise 'released_year' and 'released_at' columns to numeric and comparable formats.
- remove empty spaces and 'N/A' values.
- convert 'released_at' to separate year/month/day columns using string splitting.
- clean 'IMDb rating', 'Meta score', and 'award' columns; split 'award' column into lists.
- get average, max, min (basic statistical elements) by using pandas library.
- perform correlation analysis and other statistical tests using numpy and pandas as needed.

plotting function

- plot those relationships of imdb rating, meta score, and runtime out by using seaborn and matplotlib, for instance using sns.relplot().
- create bar plots and catplots to show the distribution of movies per genre over time (10-year intervals)- label x as years and label y as counts of genre, sns.catplot().
- use sns.lmplot() to show the imdb rating changes over time (check if the rating is growing over time).

Code & Visualization

```
In [32]: !pip install pandas -q
!pip install numpy -q
!pip install seaborn -q
!pip install matplotlib -q
!pip install scipy -q
!pip install plotly -q
!pip install "spacy==3.4.4" -q
!pip install "pydantic<2" -q
!pip install nltk -q
!pip install textblob -q
```

```
[notice] A new release of pip is available: 25.0.1 -> 25.3
[notice] To update, run: pip install --upgrade pip
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[notice] A new release of pip is available: 25.0.1 -> 25.3
[notice] To update, run: pip install --upgrade pip
```

```
In [33]: !python -m spacy download en_core_web_sm -q
```

[notice] A new release of pip is available: 25.0.1 -> 25.3
[notice] To update, run: pip install --upgrade pip
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')

```
In [35]: import pandas as pd
import seaborn as sns
import io
import numpy as np
import re
import matplotlib
import matplotlib.pyplot as plt
from scipy.stats import pearsonr, norm
from decimal import Decimal, getcontext
import plotly.express as px
import plotly.io as pio

import spacy
from textblob import TextBlob
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import contextlib

nltk_packages = ["punkt", "wordnet", "stopwords"]

for pkg in nltk_packages:
    with contextlib.redirect_stdout(None):
        nltk.download(pkg, quiet=True)
    print(f"✓ {pkg} ready")

nlp = spacy.load("en_core_web_sm")
print("spaCy loaded ✓")
```

✓ punkt ready
✓ wordnet ready
✓ stopwords ready
spaCy loaded ✓

```
In [36]: # run disney data
Disney_movies = pd.read_csv("disney_plus_shows.csv", index_col="title")

# drop unusable columns
col_to_drop = ["rated", "writer", "actors", "director"]
data_Disney = Disney_movies.drop(columns=col_to_drop)

# my test samples (without plot column) contains edge cases
my_test1 = pd.read_csv(io.StringIO(
    "title,imdb_id,type,year,released_at,added_at,runtime,genre,language,"
    "country,awards,metascore,imdb_rating,imdb_votes\n"

    "\"10 Things I Hate About You\",tt0147800,movie,1999,"
    "\"31 Mar 1999\", \"November 12, 2019\", \"97 min\", \"
    \"Comedy, Drama, Romance\", \"English, French\", \"USA\", \"
    \"2 wins & 13 nominations.\",70.0,7.3,\"283,945\"\n"

    "\"Zapped\",tt3544734,movie,2014,\"
    \"23 Jun 2014\", \"November 12, 2019\", \"102 min\", \"
    \"Comedy, Family, Fantasy\", \"English\", \"Canada, USA\", \"
    \"5 wins & 1 nomination.\",,5.1,\"3,750\"\n"

    "\"101 Dalmatian Street\",tt7019028,series,\"2018-\", \"
    \"25 Mar 2019\", \"February 28, 2020\", \"
    \"Animation, Comedy, Family\", \"English\", \"UK, USA, Canada\",,,6.2,\"
    \"124\"\n"
), index_col="title")

# add plot column to my_test1
my_test1["plot"] = [
    "A pretty, popular teenager can't go out on a date until her ill-tempered "
    "older sister does.",

    "When Zoey's mom remarries, Zoey finds it hard adjusting to her new life - "
    "no longer the only kid in the family.",

    "This series follows the lives of Delilah and Doug, descendants of Pongo "
    "and Perdita, and their 99 dalmatian pups at 101 Dalmatian Street, Camden "
    "Town."
]

my_test1
```

Out[36]:

	imdb_id	type	year	released_at	added_at	runtime	genre	language	country	awards	metascore
title											
10 Things I Hate About You	tt0147800	movie	1999	31 Mar 1999	November 12, 2019	97 min	Comedy, Drama, Romance	English, French	USA	2 wins & 13 nominations.	
	Zapped	tt3544734	movie	2014	23 Jun 2014	November 12, 2019	102 min	Comedy, Family, Fantasy	English	Canada, USA	5 wins & 1 nomination.
101 Dalmatian Street	tt7019028	series	2018–	25 Mar 2019	February 28, 2020	NaN	Animation, Comedy, Family	English	UK, USA, Canada		NaN

In [37]:

```
# run global1000 data
Global1000_movies = pd.read_csv("imdb_top_1000.csv", index_col="Series_Title")

# drop unusable columns
col_to_drop = ["Poster_Link", "Certificate", "Director", "Star1", "Star2", "Star3", "Star4"]
data_Global = Global1000_movies.drop(columns=col_to_drop)

# my test samples (without overview column) contains edge cases
my_test2 = pd.read_csv(io.StringIO(
    "Series_Title,Released_Year,Runtime,Genre,IMDB_Rating,Meta_score,No_of_Votes,Gross\n"

    "\"The Godfather\",1972,\"175 min\", \"Crime, Drama\", \"9.2\", \"100.0\", \"1620367\", \"134,966,411\"\n"

    "\"Lifeboat\",1944,\"97 min\", \"Drama, War\", \"7.6\", \"78.0\", \"26471\", \"\"\n"

    "\"Giant\",1956,\"201 min\", \"Drama, Western\", \"7.6\", \"84.0\", \"34075\", \"\"\n"
), index_col="Series_Title")

# add plot column to my_test1
my_test2["Overview"] = [
    "An organized crime dynasty's aging patriarch transfers control of his clandestine empire to his reluctant son.",

    "Several survivors of a torpedoed merchant ship in World War II find themselves in the same lifeboat with one of the crew members of the U-boat that sank their ship.",

    "Sprawling epic covering the life of a Texas cattle rancher and his family and associates."
]

# show
my_test2
```

Out [37]:

	Released_Year	Runtime	Genre	IMDB_Rating	Meta_score	No_of_Votes	Gross	Overview
Series_Title								
The Godfather	1972	175 min	Crime, Drama	9.2	100.0	1620367	134,966,411	An organized crime dynasty's aging patriarch t...
Lifeboat	1944	97 min	Drama, War	7.6	78.0	26471	NaN	Several survivors of a torpedoed merchant ship...
Giant	1956	201 min	Drama, Western	7.6	84.0	34075	NaN	Sprawling epic covering the life of a Texas ca...
Functions								

In [38]:

```
# function used for both dataset
# transform special column to original
def get_col(dataset, col_name, pattern):
    """
    Takes: dataset (DataFrame), col_name (str), pattern (str)
    Returns: DataFrame
    """
    df = dataset.copy()
    df[col_name] = df[col_name].astype(str).str.split(pattern)
    df = df.explode(col_name)
```

```
df[col_name] = df[col_name].str.strip()
df = df[df[col_name] != ""]
return df

# keeping color the same
def make_genre_color_map(df, genre_col):
    """
    Takes: df (DataFrame), genre_col (str)
    Returns: dict
    """
    unique_genres = sorted(df[genre_col].dropna().unique())
    palette = sns.color_palette("tab20", len(unique_genres))
    return {genre: palette[i] for i, genre in enumerate(unique_genres)}

def filtering_year(dataset, year):
    """
    Takes: dataset (DataFrame), year (str)
    Returns: DataFrame
    """
    df = dataset.copy()
    df[year] = pd.to_numeric(df[year], errors="coerce")
    df = df.dropna(subset=[year])
    df[year] = df[year].astype(int)
    return df

# function visualization 1: year vs. genre count
def plot_genre(dataset, data_name, year, genre, condition, color_map):
    """
    Takes: dataset (DataFrame), data_name (str), year (str), genre (str),
           condition (str), color_map (dict)
    Returns: seaborn.relplot
    """
    # filtering
    df = filtering_year(dataset, year)

    # condition
    if condition == "before":
        df = df[df[year] < 2000]
    elif condition == "after":
        df = df[df[year] >= 2000]

    # groupby
    genre_data = df.groupby([year, genre]).size().reset_index(name="count")

    # extract palette only for genres that appear
    palette = {g: color_map[g] for g in genre_data[genre].unique()}

    # plot
    genre_plot = sns.relplot(
        data=genre_data, x=year, y="count",
        hue=genre, kind="line",
        palette=palette, height=6, aspect=2
    )
    genre_plot.set(ylim=(0, 25))
    genre_plot.set(
        title=f"Genres of {data_name} over years ({condition} 2000)",
        xlabel="Year", ylabel="Numbers of genre"
    )
    plt.show()
    return genre_plot

# function visualization 2:
def plot_line(dataset, data_name, year, numeric, numeric_name, y_num):
    """
    Takes: dataset (DataFrame), data_name (str), year (str),
           numeric (str), numeric_name (str), y_num (float)
    Returns: seaborn.relplot
    """
    line_plot = sns.relplot(
        data=dataset, x=year, y=numeric, kind="line",
        height=6, aspect=2
    )
    line_plot.set(ylim=(0, y_num))

    years = dataset[year].dropna().astype(int)
    min_y, max_y = years.min(), years.max()
    ticks = list(range((min_y//20)*20, max_y+1, 20))
    line_plot.set(xticks=ticks)

    line_plot.set(
        title=f"{numeric_name} of {data_name} over years",
        xlabel="Year", ylabel=numeric_name
    )

    plt.show()
```

```

    return line_plot

# data transform
stop_words = set(stopwords.words('english'))
positive_words = {"great","good","fun","happy","love","exciting","enjoy","positive","charming"}
negative_words = {"bad","dark","death","hate","fear","negative","sad","danger","cruel"}

def extract_linguistic_features(df, text_col="description"):
    """
    Takes: df (DataFrame), text_col (str)
    Returns: DataFrame with added linguistic features:
            polarity, subjectivity, adj_density,
            pos_count, neg_count, sentiment_contrast
    """
    df = df.copy()

    def process(text):
        if pd.isnull(text): return pd.Series({
            "polarity": 0, "subjectivity": 0,
            "adj_density": 0,
            "pos_count": 0, "neg_count": 0,
            "sentiment_contrast": 0
        })

        blob = TextBlob(text)
        polarity = blob.sentiment.polarity
        subjectivity = blob.sentiment.subjectivity

        tokens = [w.lower() for w in word_tokenize(text) if w.isalpha()]

        # adj
        adj_count = sum(1 for w, tag in blob.tags if tag.startswith("JJ"))
        adj_density = adj_count / max(len(tokens), 1)

        pos_count = sum(1 for w in tokens if w in positive_words)
        neg_count = sum(1 for w in tokens if w in negative_words)

        sentiment_contrast = pos_count - neg_count

        return pd.Series({
            "polarity": polarity,
            "subjectivity": subjectivity,
            "adj_density": adj_density,
            "pos_count": pos_count,
            "neg_count": neg_count,
            "sentiment_contrast": sentiment_contrast
        })

    df_features = df[text_col].apply(process)
    return pd.concat([df, df_features], axis=1)

# sum/groupby (considering edgecases)
def aggregate_by_year(df, year_col="year", feature="polarity"):
    """
    Takes: df (DataFrame), year_col (str), feature (str)
    Returns: DataFrame with yearly mean of the feature
    """
    df = df.copy()
    df[year_col] = pd.to_numeric(df[year_col], errors="coerce")\
        .fillna(0).astype(int)

    yearly = (
        df.groupby(year_col)[feature]
        .mean()
        .reset_index()
        .rename(columns={feature: f"{feature}_yearly"})
    )
    return yearly

# change runtime to just int/float numbers
def runtime(dataset, col_name):
    """
    Takes: dataset (DataFrame), col_name (str)
    Returns: DataFrame
    """
    dataset[col_name] = (dataset[col_name].astype(str).str.extract(r"(\d+)").astype(float))
    return dataset

# function visualization 3
nlp = spacy.load("en_core_web_sm")

def parse_awards(text):
    """
    Takes: text (str)

```

```
Returns: Series of parsed award counts (wins/nominations per award type)
"""
if pd.isna(text):
    return pd.Series({
        "oscars_won": 0, "oscars_nominated": 0,
        "globes_won": 0, "globes_nominated": 0,
        "emmys_won": 0, "emmys_nominated": 0,
        "bafta_won": 0, "bafta_nominated": 0,
        "sag_won": 0, "sag_nominated": 0,
        "total_wins": 0, "total_nominations": 0
    })

text = text.lower()

# extract award counts
def extract(pattern):
    nums = re.findall(pattern, text)
    return sum(int(n) for n in nums) if nums else 0

# specific major awards
oscars_won = extract(r"won\s+(\d+)\s+oscar")
oscars_nominated = extract(r"nominated\s+for\s+(\d+)\s+oscar")

globes_won = extract(r"won\s+(\d+)\s+golden globe")
globes_nominated = extract(r"nominated\s+for\s+(\d+)\s+golden globe")

emmys_won = extract(r"won\s+(\d+)\s+.*?emmy")
emmys_nominated = extract(r"nominated\s+for\s+(\d+)\s+.*?emmy")

bafta_won = extract(r"won\s+(\d+)\s+bafta")
bafta_nominated = extract(r"nominated\s+for\s+(\d+)\s+bafta")

sag_won = extract(r"won\s+(\d+)\s+sag")
sag_nominated = extract(r"nominated\s+for\s+(\d+)\s+sag")

# generic wins / nominations
wins = extract(r"(\d+)\s+win")
nominations = extract(r"(\d+)\s+nomination")

# another wins
another_wins = extract(r"another\s+(\d+)\s+win")
another_nominations = extract(r"another\s+(\d+)\s+nomination")

return pd.Series({
    "oscars_won": oscars_won,
    "oscars_nominated": oscars_nominated,

    "globes_won": globes_won,
    "globes_nominated": globes_nominated,

    "emmys_won": emmys_won,
    "emmys_nominated": emmys_nominated,

    "bafta_won": bafta_won,
    "bafta_nominated": bafta_nominated,

    "sag_won": sag_won,
    "sag_nominated": sag_nominated,

    # Combine generic wins + another wins
    "total_wins": wins + another_wins,
    "total_nominations": nominations + another_nominations
})

def plot_scatter(dataset, x_name, y_name, data_name):
    """
    Takes: dataset (DataFrame), x_name (str), y_name (str), data_name (str)
    Returns: seaborn.lmplot
    """
    relat_data = dataset.loc[:, [x_name, y_name]]
    relat_plot = sns.lmplot(relat_data, x=x_name, y=y_name, height=6, aspect=3)
    relat_plot.set(title=f"The relationship between {x_name} and {y_name} in {data_name} shows",
                   xlabel=f"{x_name} of movies & series", ylabel=f"{y_name} of movies & series")
    plt.show()
    return relat_plot

def plot_bar(dataset, x_name, y_name, data_name, y_ini, y_step):
    """
    Takes: dataset (DataFrame), x_name (str), y_name (str),
           data_name (str), y_ini (float), y_step (float)
    Returns: seaborn.catplot
    """
    bar_plot = sns.catplot(data=dataset, x=x_name, y=y_name, kind="bar", height=6, aspect=2)
    ax = bar_plot.ax
    ax.set_ylim(y_ini, 9)
    ax.set_yticks(np.arange(y_ini, 8.5, y_step))
    ax.tick_params(axis="x", rotation=60)
    ax.set_title(f"The relationship between {x_name} and {y_name} in {data_name} shows")
```

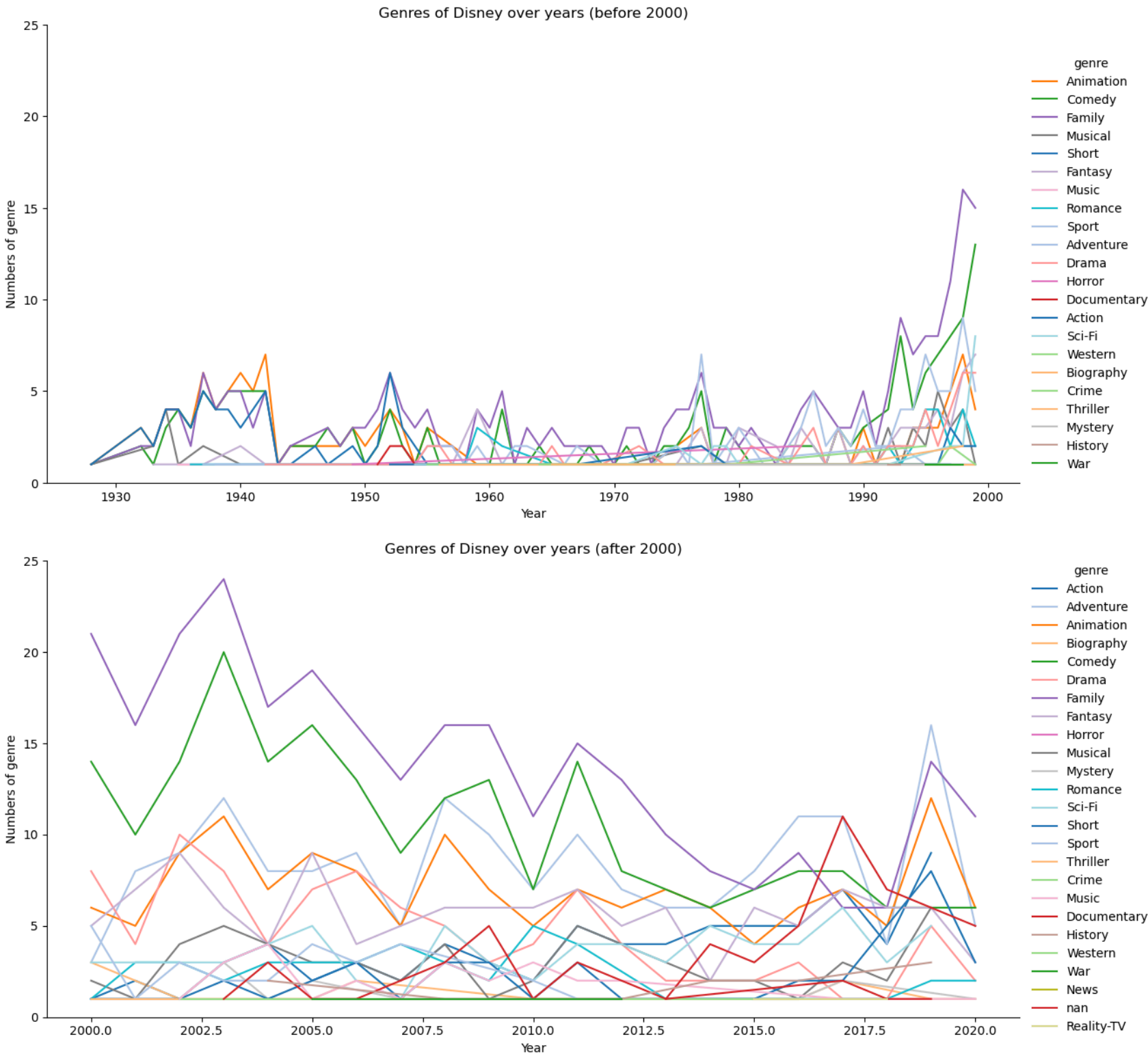
```
ax.set_xlabel(f"{x_name} of movies & series")
ax.set_ylabel(f"{y_name} of movies & series")
plt.show()
return bar_plot
```

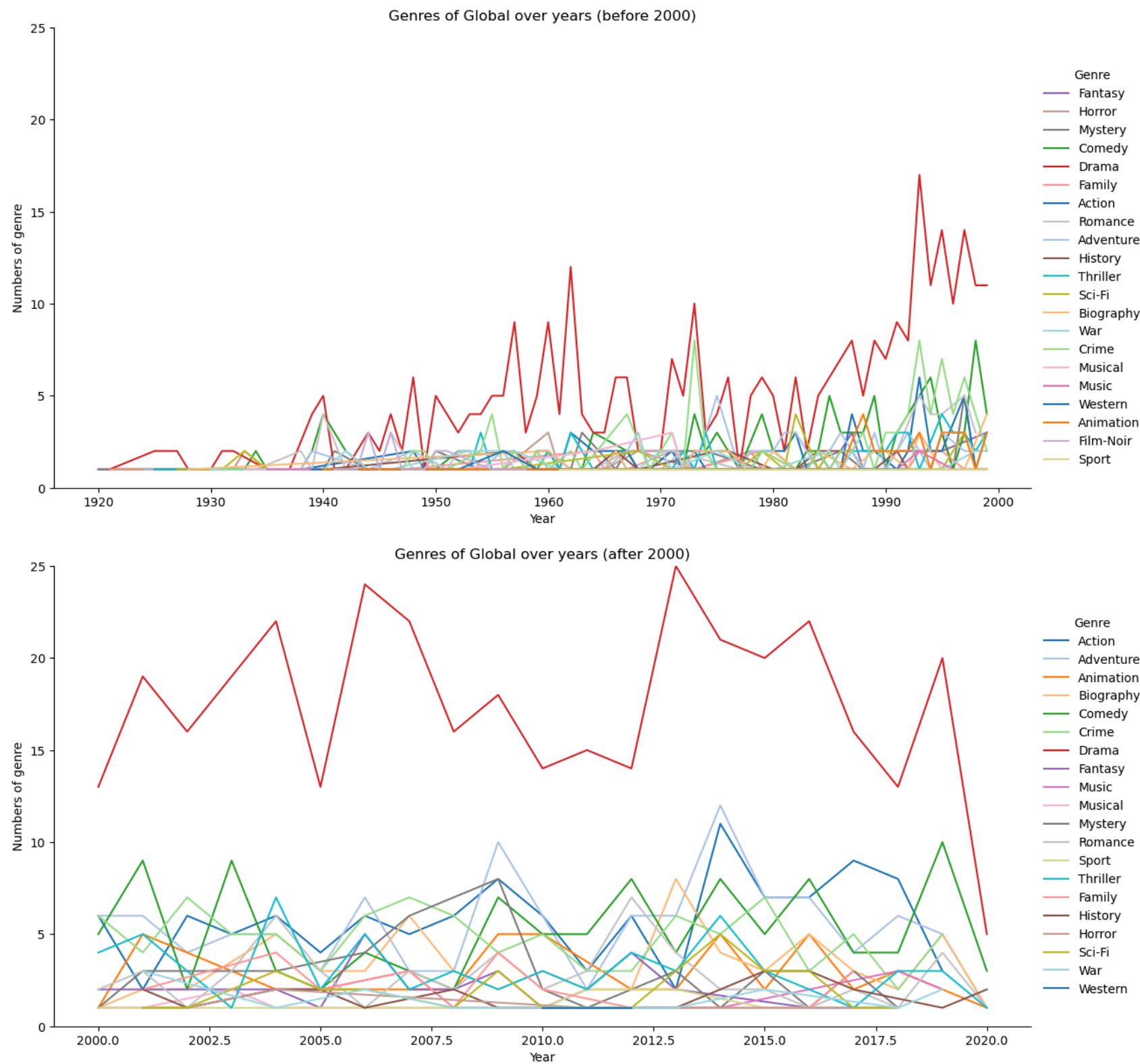
```
In [39]: # visualization 1: year vs. genre counts (normal visualization)
# disney before & after 2000
data_Disney = get_col(data_Disney, "genre", ",")
DISNEY_GENRE_COLOR_MAP = make_genre_color_map(data_Disney, "genre")
disney_genre_before = plot_genre(data_Disney, "Disney", "year", "genre", "before",
                                DISNEY_GENRE_COLOR_MAP)

disney_genre_after = plot_genre(data_Disney, "Disney", "year", "genre", "after",
                                DISNEY_GENRE_COLOR_MAP)

# global before & after 2000
data_Global = get_col(data_Global, "Genre", ",")
GLOBAL_GENRE_COLOR_MAP = make_genre_color_map(data_Global, "Genre")
global_genre_before = plot_genre(data_Global, "Global", "Released_Year", "Genre", "before",
                                GLOBAL_GENRE_COLOR_MAP)

global_genre_after = plot_genre(data_Global, "Global", "Released_Year", "Genre", "after",
                                GLOBAL_GENRE_COLOR_MAP)
```





Visualization 1 (sns plot)

Disney dataset:

Caption: As shown in both the “before 2000” and “after 2000” line plots, Disney films reached their highest point in the family-oriented genre around 1997 and have maintained the broadest range of genres ever since.

Alt Text: A dense multi-line plot showing the number of Disney films in various genres from 2000 to 2021. Genre activity is higher and lines fluctuate more compared to the pre-2000 plot, with frequent peaks in Animation, Family, Action, Fantasy, and Adventure around the mid-2010s, followed by a clear decline near 2020.

Brief Explanation & Analysis

This plot displays the distribution of Disney film genres from 2000 to 2021. Each colored line represents one genre and tracks how many films of that genre were released in each year. I selected a line chart because it is ideal for showing multiple long-term trends simultaneously and helps readers compare changes across genres. From this plot, readers should recognize a major shift: after 2000, Disney not only released significantly more films per year but also embraced a wider range of genres, including action, adventure, fantasy, and thrillers. The early 21st century shows steady growth, peaking between 2015–2017 across many genres. A sharp decline appears around 2019–2020, possibly due to the COVID-19 pandemic halting film production. In summary, the post-2000 period reflects both expansion and diversification in Disney’s film output.

Comparing the two plots, we can see that Disney’s production before 2000 was modest and concentrated in a few core genres—mainly family, comedy, and animation. After 2000, Disney expanded sharply in both quantity and diversity, producing more films across action, adventure, fantasy, and other genres. However, the number of family films has remained almost the highest almost all the time. The film output reached its highest point in the mid-2010s, followed by a noticeable drop around 2020. Overall, Disney’s catalog transformed from a narrow genre focus to a far more diverse and high-volume production landscape.

Global dataset:

Caption: Across both the “before 2000” and “after 2000” plots, the global film industry remains dominated by Family, Drama, and Comedy, but the scale and diversity of production increase sharply after 2000. The post-2000 era shows a dramatic expansion in genre variety and overall film volume, reaching broad peaks in the mid-2010s before dropping around 2020.

Alt Text: A pair of multi-line plots charting the number of global films released in each genre over time. In the pre-2000 plot, most genre lines remain very low and steady from the 1920s through the 1980s, with only small spikes in Family, Drama, and Comedy. In the post-2000 plot, the lines become much denser and fluctuate widely, with genres like Action, Adventure, Comedy, Animation, and Fantasy rising sharply in the mid-2010s before declining noticeably near 2020.

Brief Explanation & Analysis

This visualization presents how global film production has changed over time, divided into two periods: before 2000 and after 2000. Each line corresponds to a specific genre, and the vertical axis shows the yearly count of films released in that category. A multi-line plot is used to highlight long-term trajectories and allow direct comparison of how genre activity evolves across decades.

In the pre-2000 panel, global filmmaking appears relatively small in scale and concentrated in a few core genres. Only categories like Family, Drama, and Comedy show occasional growth, while most other genres remain minimal until the late 20th century. Overall production increases slowly, with limited fluctuations across decades. The post-2000 panel, however, displays a marked shift. The global industry expands rapidly, producing far more films per year and engaging in a much wider range of genres. Categories such as Action, Adventure, Animation, Sci-Fi, and Thriller show substantial growth, with many reaching peak activity in the mid-2010s. A clear decline emerges around 2019–2020, likely reflecting the worldwide disruption caused by the COVID-19 pandemic.

Taken together, the two plots illustrate a major transformation: the global film industry evolves from a low-volume, genre-restricted landscape into a highly diverse and high-output system, while Family, Drama, and Comedy continue to serve as consistently strong contributors across both time periods.

```
In [40]: # visualization 1: year vs. genre counts (special visualization-- plotly)
# disney
genre_year_df = (
    data_Disney
    .groupby(["year", "genre"])
    .size()
    .reset_index(name="count")
)

genre_year_disney = filtering_year(genre_year_df, "year")
fig_disney = px.line(
    genre_year_disney,
    x="year",
    y="count",
    color="genre",
    title="Disney Genre Trends Over Time (Interactive)",
    markers=True
)

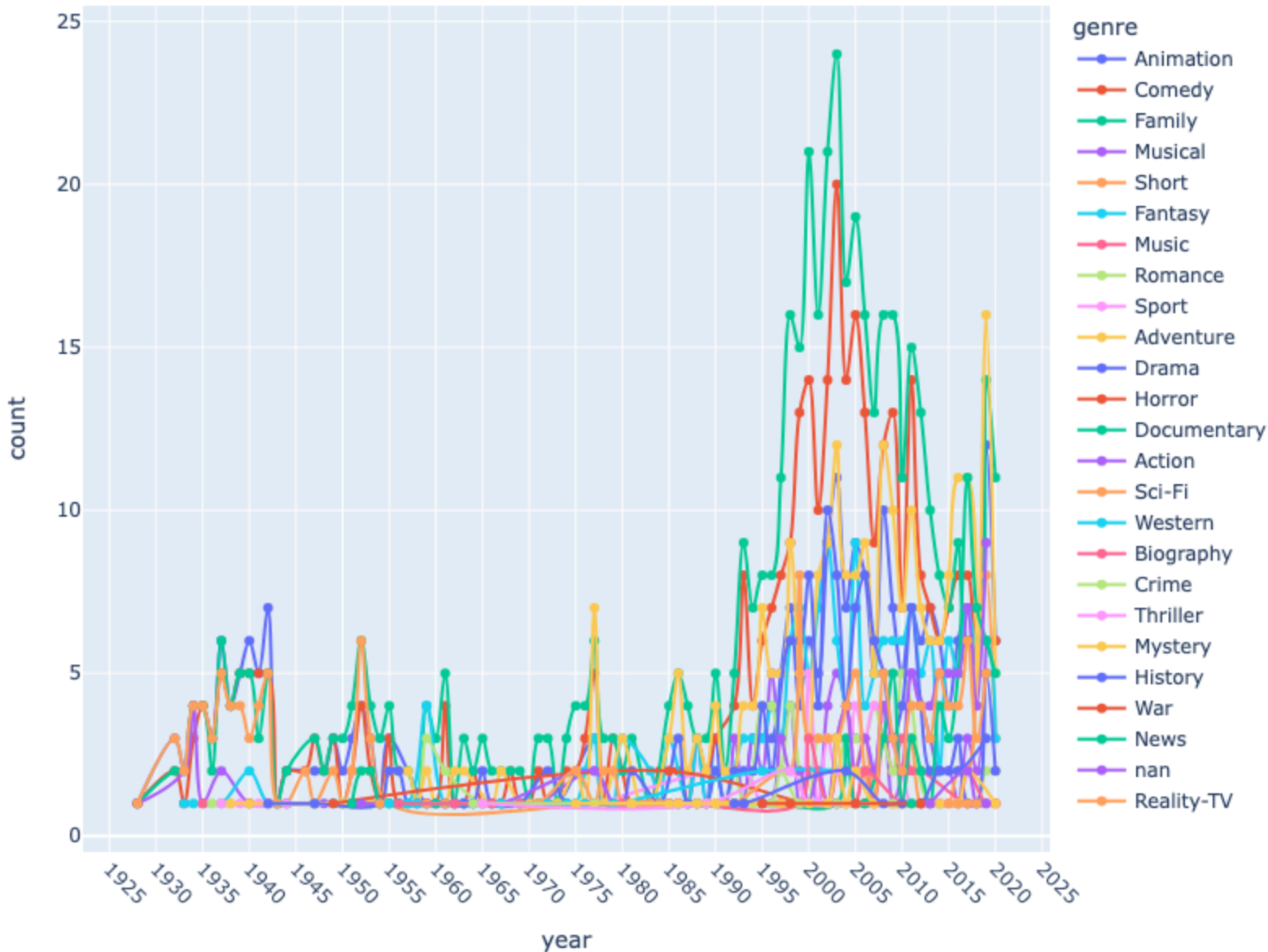
# in one plot
fig_disney.update_xaxes(
    tickangle=45,
    tickmode="linear",
    dtick=5
)

fig_disney.update_layout(
    hovermode="x unified",
    width=1100,
    height=700
)

fig_disney.update_traces(line_shape="spline")

# for me to visualize
# fig_disney.show()
```

Disney Genre Trends Over Time (Interactive)



For interactive plot cannot shown in pdf, here is the picture of it Disney Genre Trends Over Time (Interactive)

```
In [41]: # global
genre_year_df_1 = (
    data_Global
    .groupby(["Released_Year", "Genre"])
    .size()
    .reset_index(name="count")
)

genre_year_global = filtering_year(genre_year_df_1, "Released_Year")
fig_global = px.line(
    genre_year_global,
    x="Released_Year",
    y="count",
    color="Genre",
    title="Global Genre Trends Over Time (Interactive)",
    markers=True
)

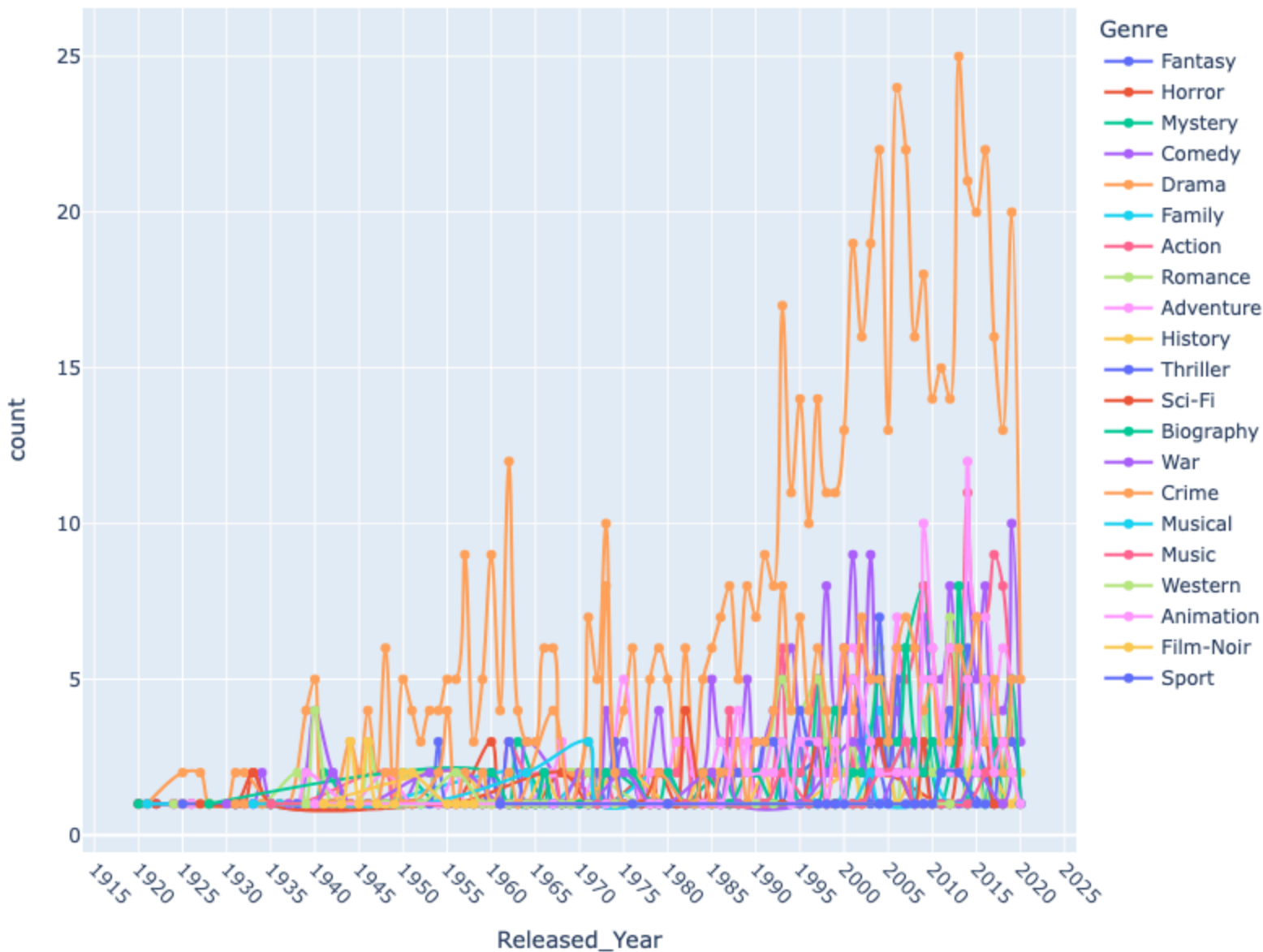
# in one plot
fig_global.update_xaxes(
    tickangle=45,
    tickmode="linear",
    dtick=5
)

fig_global.update_layout(
    hovermode="x unified",
    width=1100,
    height=700
)

fig_global.update_traces(line_shape="spline")

# for me to visualize
# fig_global.show()
```

Global Genre Trends Over Time (Interactive)



For interactive plot cannot shown in pdf, here is the picture of it Global Genre Trends Over Time (Interactive)

Visualization 1 (plotly plot)

Disney dataset:

Caption: This interactive line plot shows how the popularity and frequency of different Disney film genres have shifted over time, highlighting a major expansion in genre diversity and production volume after the 1990s.

Alt Text: This interactive plot displays multiple colored lines representing different Disney movie genres from 1925 to 2025. The x-axis shows the release year, and the y-axis shows how many movies in each genre were produced that year. Each genre has its own colored line, forming waves or spikes across the timeline. The early decades show very low and stable counts, while from the 1990s onward, many genres rise sharply with noticeable peaks around the early 2000s. A legend on the right lists all genres, such as Animation, Comedy, Family, Fantasy, Drama, Adventure, Sci-Fi, and more. When viewed interactively, hovering over points reveals exact counts and years for each genre.

Brief Explanation & Analysis

This interactive plot visualizes how Disney's genre production evolved over nearly a century by displaying the count of films per genre per year. I selected this plot because an interactive multi-line time series allows readers to explore individual genres dynamically—hovering over points helps isolate patterns within specific categories despite the visual complexity of many overlapping lines. The plot reveals that Disney produced relatively few films before 1980 across all genres, but output expanded dramatically in the 1990s and early 2000s, particularly in Family, Comedy, Adventure, and Fantasy genres. The visible spikes indicate periods when Disney diversified its content in response to industry trends and technological advances. The key takeaway is that Disney's genre landscape became significantly broader and more active over time, with genre production accelerating sharply in the modern era.

Global dataset:

Caption: This interactive plot illustrates how global film genres have fluctuated in popularity over the past century, with Comedy consistently dominating production volume compared to other categories.

Alt Text: This interactive multi-line plot shows film genre trends worldwide from roughly 1915 to 2025. The horizontal axis represents release years, and the vertical axis displays the number of films produced each year for each genre. Many thin, overlapping colored lines represent genres such as Fantasy, Horror, Mystery, Comedy, Drama, Family, Action, Romance, Sci-Fi, Western, Animation, and others. The orange Comedy line stands out as the highest and most variable, rising sharply after the 1980s and peaking above 20 films per year in the early 2000s. Other genres show much lower but gradually increasing counts, with occasional spikes in Adventure, Family, and Thriller. The interactive version allows users to hover over points to reveal exact values, making it easier to isolate individual genre trends within the dense cluster of lines.

Brief Explanation & Analysis

This interactive plot visualizes the relationship between film genre and release year, showing how many movies of each genre were produced annually on a global scale. I selected this interactive line plot because it enables users to explore detailed patterns that would otherwise be visually overwhelming—hovering over individual data points helps separate densely overlapping lines and highlights subtle rises or declines in genre popularity. The most striking pattern is that Comedy consistently shows the highest production volume, especially from the 1990s onward, suggesting strong global demand for comedic films. Other genres such as Adventure, Thriller, and Family films show moderate growth but remain far below Comedy's production peaks. The key takeaway from this plot is that while global filmmaking diversified significantly over the last century, Comedy remained the most reliably produced and widely popular genre across time.

```
In [42]: # visualization 2
# year vs. imdb votes
# disney
data_Disney["imdb_votes"] = (data_Disney["imdb_votes"].astype(str).str.replace(",", "",
                                                                    regex=False)
                             .astype(float))
vote_Disney = (filtering_year(data_Disney, "year").groupby("year", as_index=False)["imdb_votes"]
               .sum())
disney_vote = plot_line(vote_Disney, "Disney", "year", "imdb_votes", "Gross imdb votes", 1e7+3e6)

# global
vote_Global = (filtering_year(data_Global, "Released_Year").groupby("Released_Year",
                                                                    as_index=False)
               ["No_of_Votes"].sum())
global_vote = plot_line(vote_Global, "Global", "Released_Year", "No_of_Votes", "Gross imdb votes",
                        3e7+5e6)

# year vs. imdb rating
# disney
rating_Disney = (filtering_year(data_Disney, "year").groupby("year", as_index=False)["imdb_rating"]
                 .mean())
disney_rating = plot_line(rating_Disney, "Disney", "year", "imdb_rating", "Average imdb rating", 10)

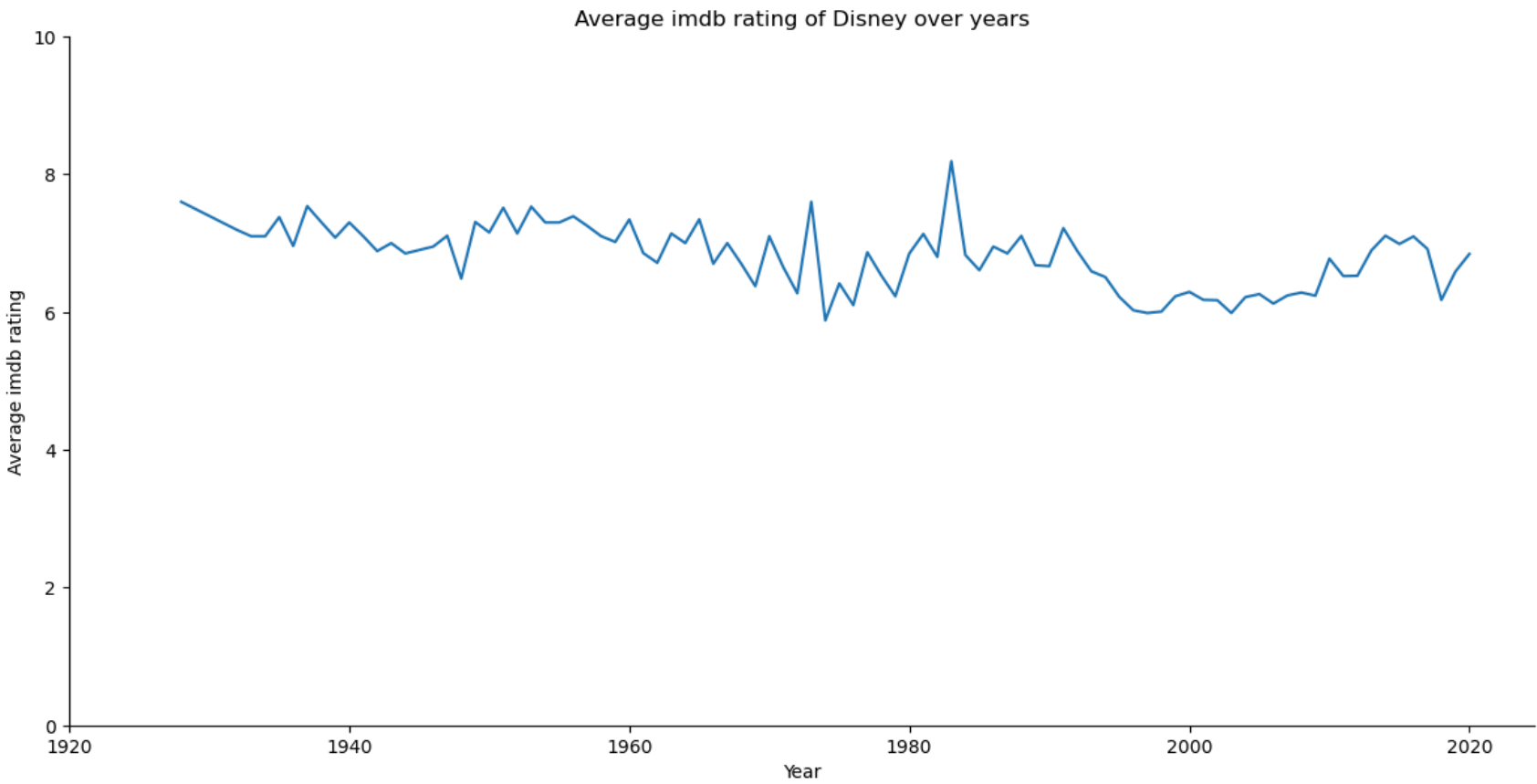
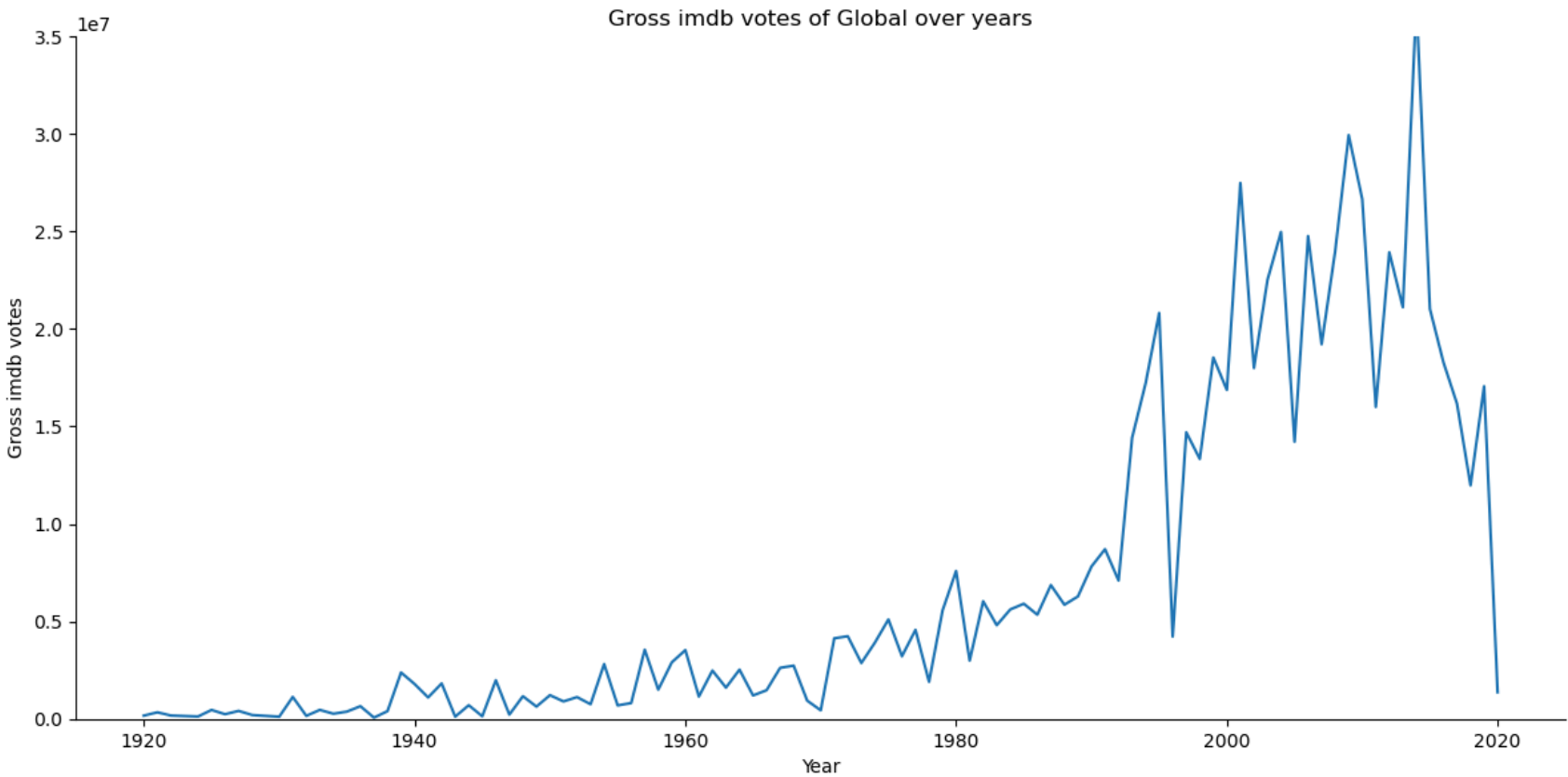
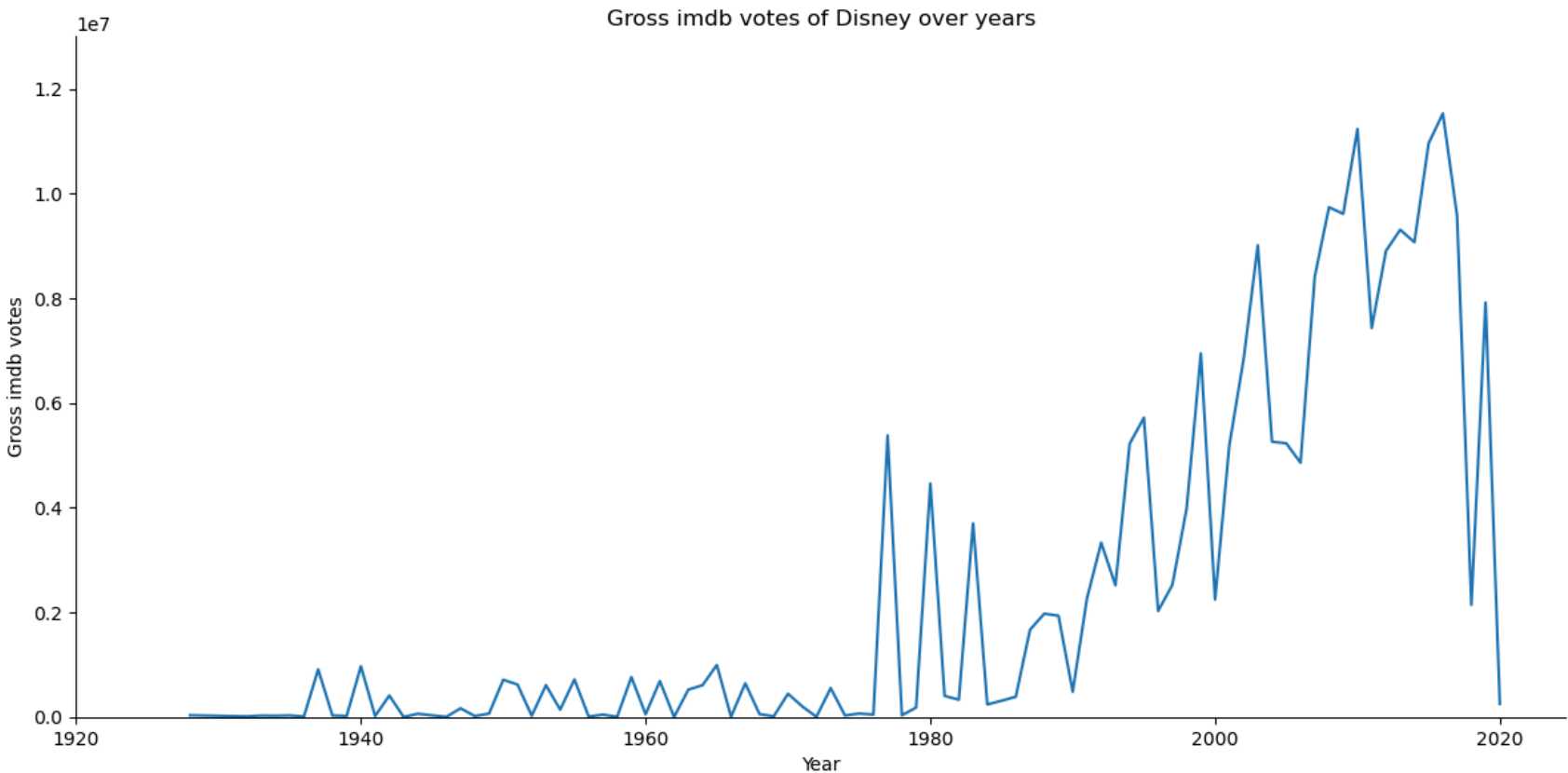
# global
rating_Global = (filtering_year(data_Global, "Released_Year").groupby("Released_Year",
                                                                    as_index=False)
                 ["IMDB_Rating"].mean())
global_rating = plot_line(rating_Global, "Global", "Released_Year", "IMDB_Rating", "Average imdb rating",
                          10)

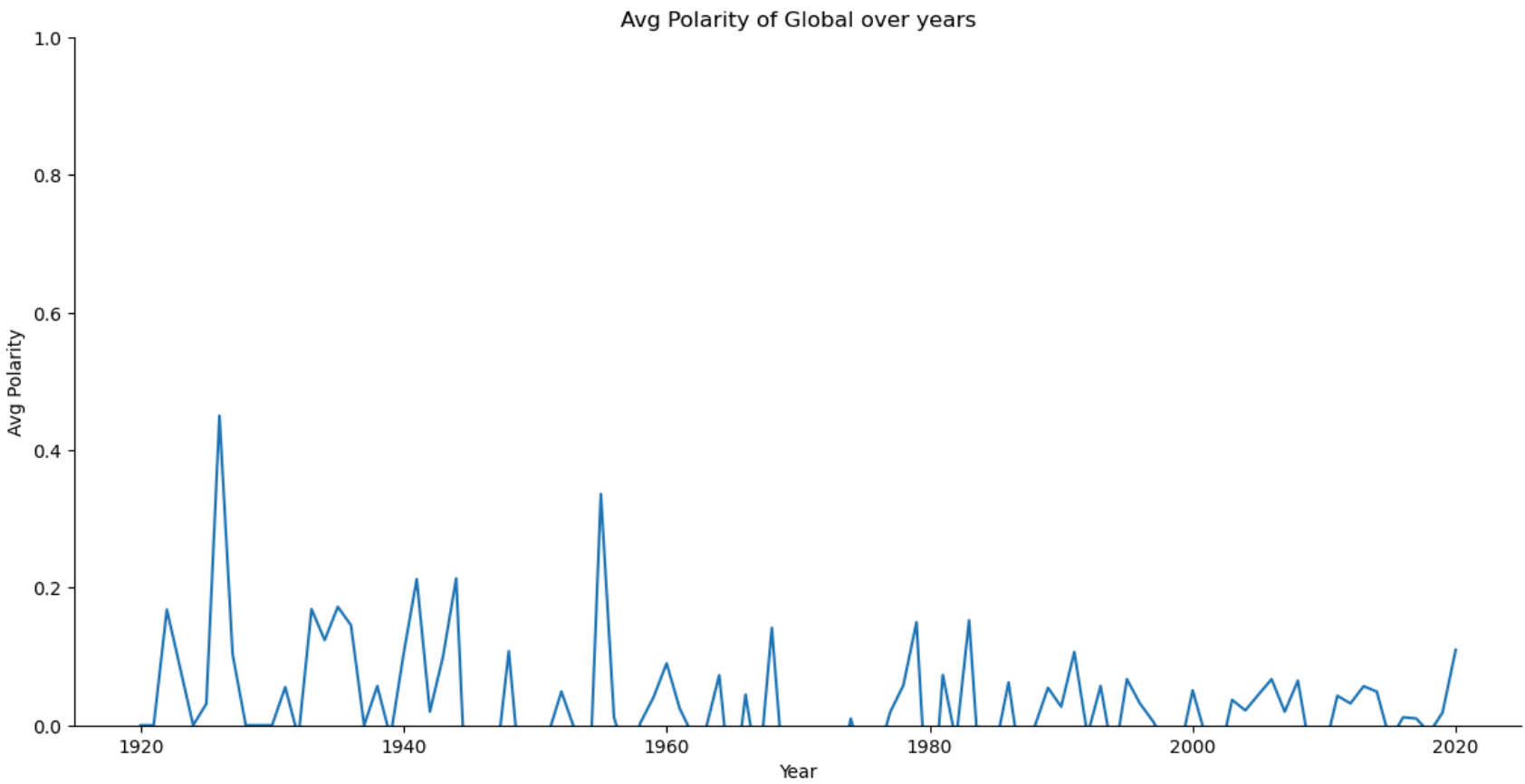
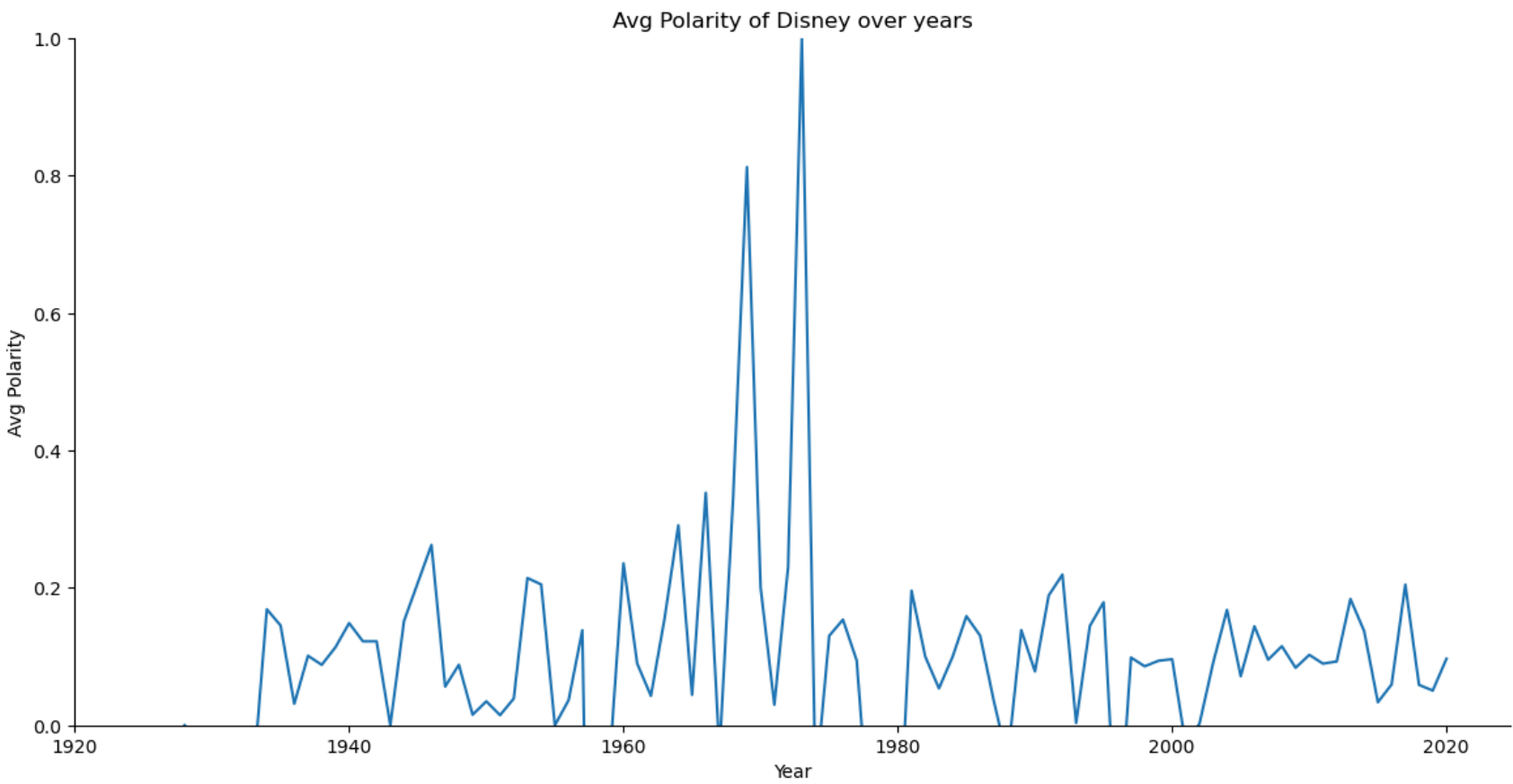
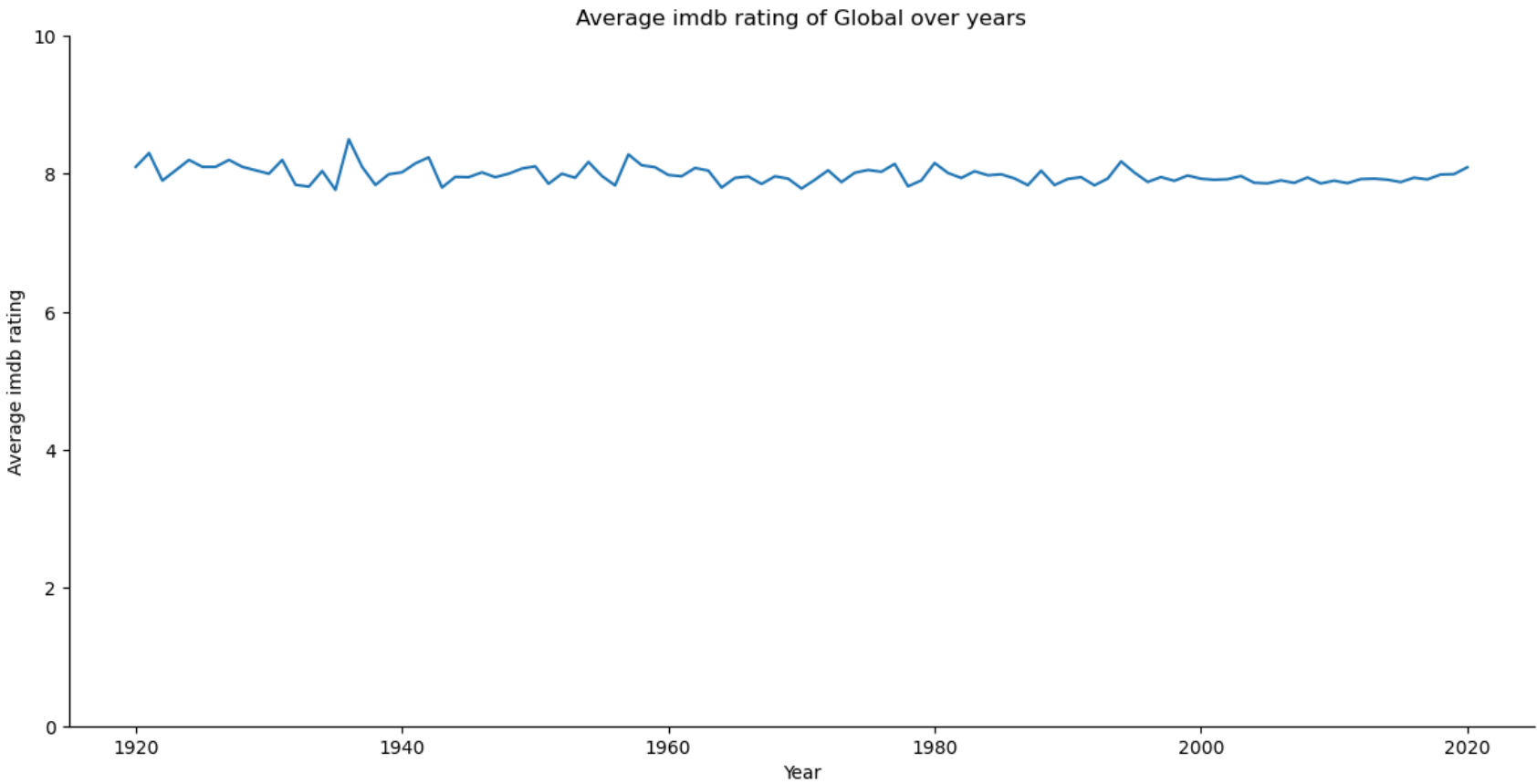
# year vs. overview (linguistic features)
# disney
data_Disney = filtering_year(data_Disney, "year")
Disney_features = extract_linguistic_features(data_Disney, text_col="plot")
Disney_yearly = aggregate_by_year(Disney_features, year_col="year", feature="polarity")

plot_line(
    dataset=Disney_yearly,
    data_name="Disney",
    year="year",
    numeric="polarity_yearly",
    numeric_name="Avg Polarity",
    y_num=1
)

# global
data_Global = filtering_year(data_Global, "Released_Year")
Global_features = extract_linguistic_features(data_Global, text_col="Overview")
Global_yearly = aggregate_by_year(Global_features, year_col="Released_Year", feature="polarity")

plot_line(
    dataset=Global_yearly,
    data_name="Global",
    year="Released_Year",
    numeric="polarity_yearly",
    numeric_name="Avg Polarity",
    y_num=1
)
```





Out[42]: <seaborn.axisgrid.FacetGrid at 0x7fef39511e10>

Visualization 2

Gross imdb votes (Disney vs. Global)

Caption: These two plots show that both global films and Disney films experienced sharp increases in audience engagement (as measured by IMDb votes) after 1980, with Disney's rise occurring later and at a smaller scale, and both showing a dramatic decline around 2020.

Alt Text: Two single-line graphs showing the yearly total IMDb votes for global films and for Disney films. In the global plot, IMDb votes remain very low from the 1920s to the 1970s, begin rising gradually in the 1980s, and then spike dramatically from the 1990s to the mid-2010s before dropping sharply around 2020. In the Disney plot, audience votes are almost negligible before 1980, increase sharply starting in the 1990s, peak in the late 2000s and mid-2010s, and similarly fall steeply near 2020.

Brief Explanation & Analysis

These two plots display the yearly total number of IMDb audience votes for global films and Disney films. IMDb votes serve as a measure of audience engagement or popularity. A line plot is used because it clearly highlights long-term trends and major shifts across decades.

From the global plot, we observe that audience engagement remained minimal for most of the 20th century, but began accelerating sharply in the 1980s and especially in the 1990s and 2000s, reaching its highest levels between 2010 and 2017. A sharp decline appears around 2020, likely due to the COVID-19 impact on film releases and viewership dynamics. The Disney plot shows a similar pattern but on a different scale. Disney films receive relatively few votes before the 1980s, followed by rapid growth beginning in the mid-1990s and a pronounced rise through the 2000s and 2010s. The magnitude of Disney’s vote count is consistently lower than that of global films, reflecting the company’s smaller share of overall film production. Like the global industry, Disney experiences a steep decline after 2019.

Taken together, the two plots illustrate parallel trends: both global and Disney films saw dramatic increases in audience engagement in the modern era, with peaks in the 2010s and a sudden downturn around 2020. The comparison highlights how Disney’s popularity growth followed the broader global trend but at a smaller scale and with slightly later timing.

Average imdb rating (Disney vs. Global)

Caption: These two plots show that global film ratings have remained consistently high and stable over time, while Disney’s ratings exhibit more variation and a slight downward trend in the late 20th and early 21st century.

Alt Text: Two single-line charts comparing the average IMDb ratings of global films and Disney films from the early 1900s to 2023. The global plot shows a nearly flat line around 8.0–8.3 for over a century, with only small fluctuations. The Disney plot shows more noticeable ups and downs between roughly 6.5 and 7.8, with higher values around the mid-20th century, a decline around 1990–2010, and a mild recovery approaching 2020.

Brief Explanation & Analysis

This visualization compares the average IMDb rating of global films (Plot A) with those of Disney films (Plot B) across the past century. IMDb ratings serve as a measure of audience appreciation or perceived quality. A line plot is used because it clearly illustrates long-term changes and highlights contrasts between two time-dependent variables.

The global film rating plot shows remarkable stability, with average ratings consistently hovering around 8.0 for nearly 100 years. This suggests that—even as production volume, audience size, and cinematic trends changed—viewer-perceived quality of globally popular films did not shift dramatically. The Disney rating plot, however, shows more fluctuation. Disney films reached relatively high ratings (around 7.5–7.8) in earlier decades but experienced a gradual decline in the late 20th and early 21st century, followed by a slight rebound approaching 2020. The narrower vertical scale indicates that Disney films generally receive lower ratings than global averages, and their audience perception varies more over time.

Taken together, the visualization highlights a meaningful contrast: global film ratings remain stable and consistently high, while Disney films show more volatility and a modest downward trend before stabilizing again. This suggests that Disney’s audience reception is more sensitive to changing production strategies, genre shifts, and franchise evolution, whereas globally popular films maintain a steadier level of viewer approval.

Avg Polarity (Disney vs. Global)

Caption: These plots show that Disney movie plot descriptions display much stronger emotional variation over time, while global movie overviews remain consistently neutral with minimal polarity fluctuation.

Alt Text: This figure contains two line charts comparing the average yearly polarity of Disney and global movies from the early 1900s to the 2020s. The Disney line fluctuates widely near zero with several sharp spikes rising above 0.8–1.0 between the 1960s and 1970s, followed by more moderate but still irregular variations after 1980. The global movie line stays close to zero across the entire timeline, with small sporadic peaks around early decades but no values approaching the larger Disney spikes. Both series show polarity values mostly below 0.2 in recent decades. The y-axis ranges from 0 to 1.0, and the x-axis spans roughly 1920–2023.

Brief Explanation & Analysis

These plots visualize the average yearly polarity extracted from text descriptions—Disney’s plot summaries and global films’ overview descriptions—using TextBlob sentiment analysis. Polarity measures the emotional tone of the text (from negative to positive), making it a linguistic indicator of how movie narratives are framed over time. A line plot is appropriate because it highlights long-term temporal changes and reveals differences between the two datasets’ emotional patterns.

The results show a clear contrast. Global movies maintain consistently neutral polarity for more than a century, rarely deviating far from zero. This indicates that global film overviews tend to use emotionally neutral and informational language, even as film trends evolve. Disney movies, however, exhibit much greater variability. The large polarity spikes in the 1960s–

1970s suggest periods where Disney plot summaries were written in unusually positive or emotionally charged language. Although polarity stabilizes somewhat after 1980, Disney still shows more fluctuation than the global dataset.

Together, these patterns provide insight into your research question about the evolution of audience perception and popularity when incorporating linguistic cues. The greater emotional expressiveness in Disney’s plot descriptions aligns with Disney’s branding—optimistic storytelling and family-oriented narratives—which differentiates it from the more neutral global film market. These linguistic differences complement numerical indicators (IMDb ratings and votes) by showing that Disney’s narrative tone is uniquely positive and varies more sharply across decades, reflecting shifts in production strategies and audience targeting.

```
In [43]: # visualization 3
# metascore vs. imdb rating
# disney
disney_meta = plot_scatter(data_Disney, "metascore", "imdb_rating", "Disney")

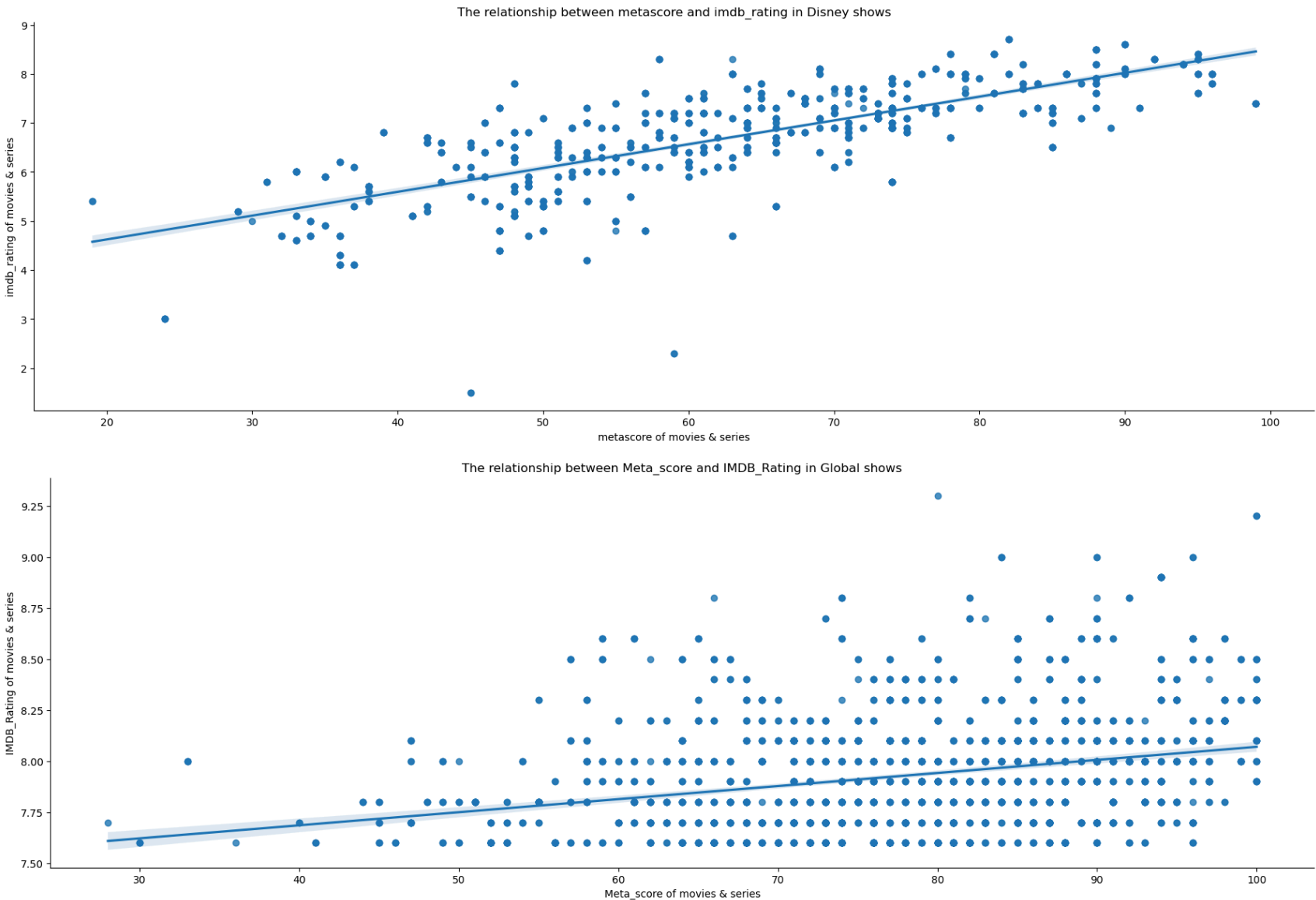
# global
global_meta = plot_scatter(data_Global, "Meta_score", "IMDB_Rating", "Global")

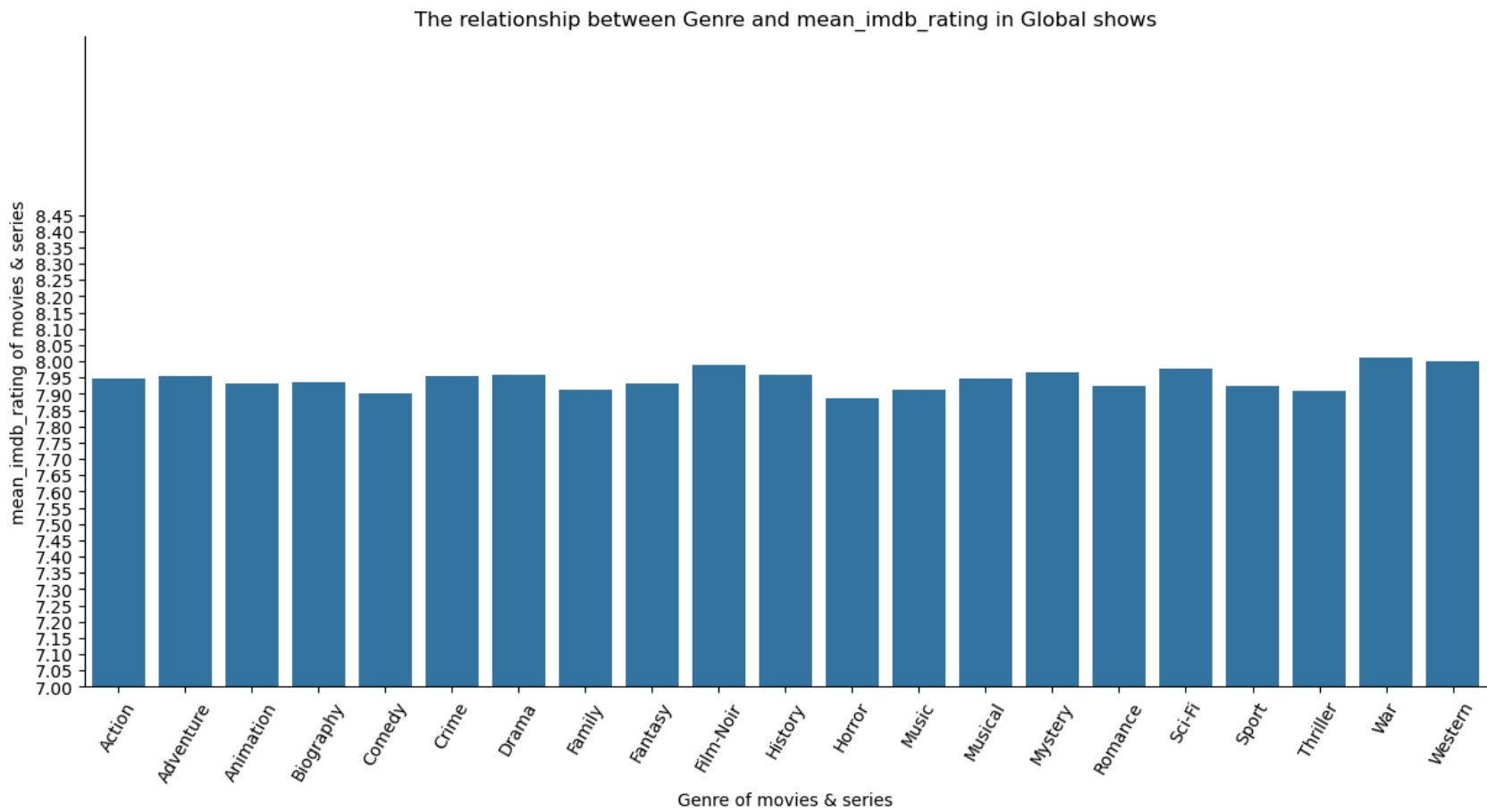
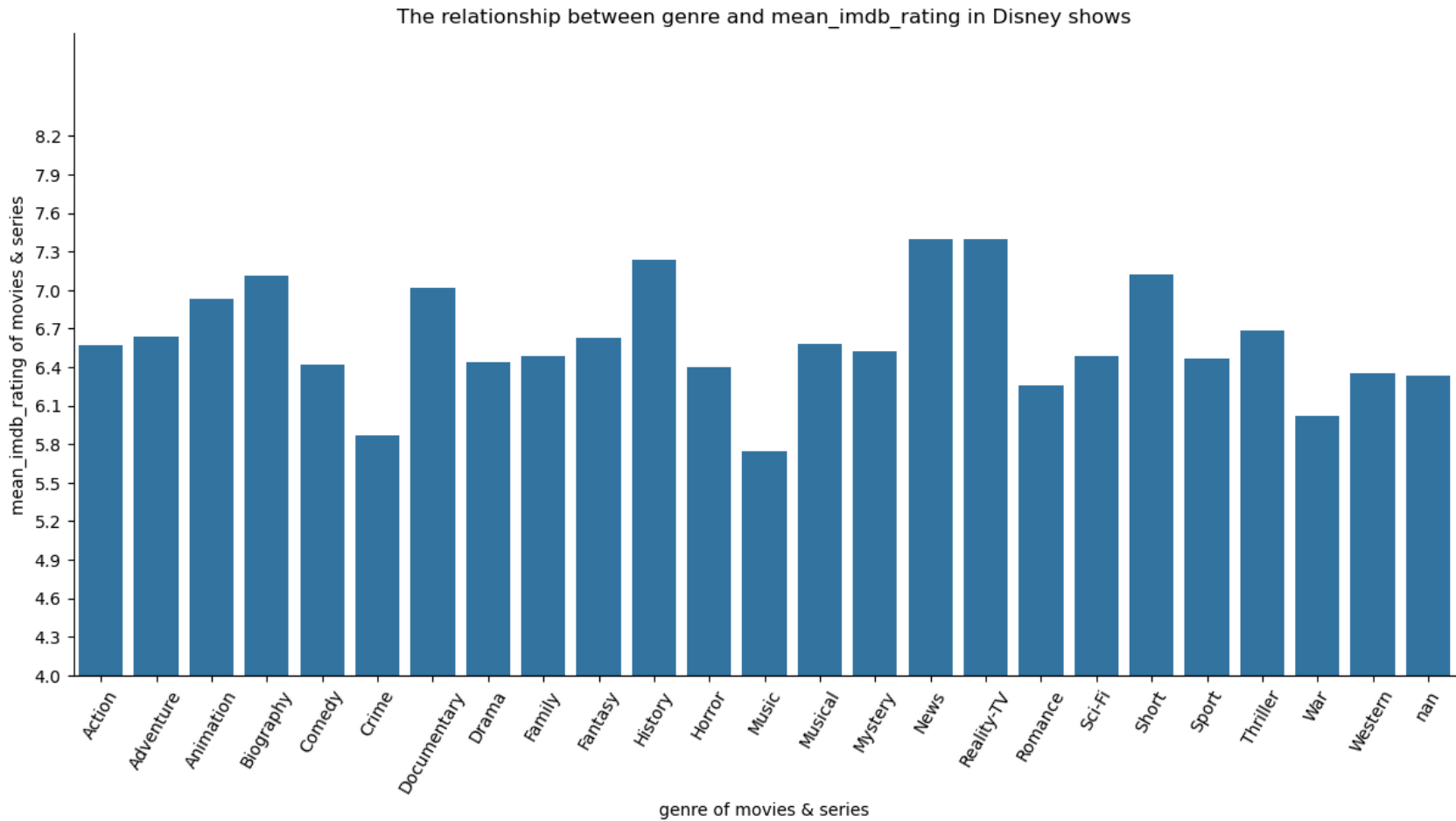
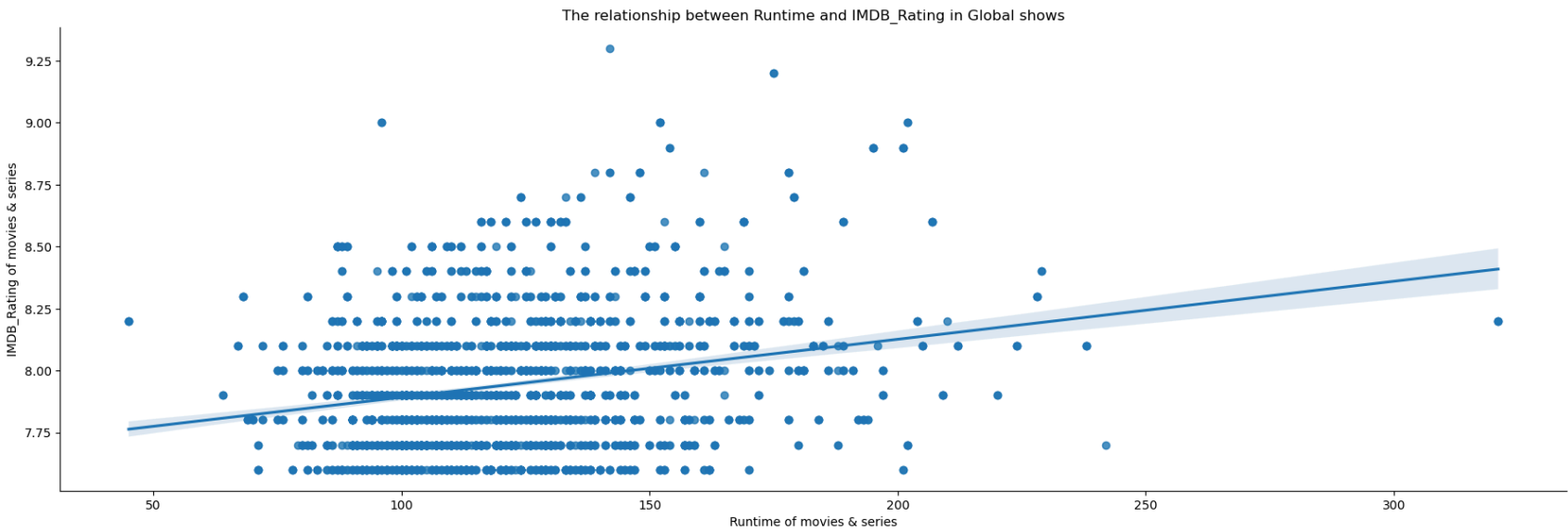
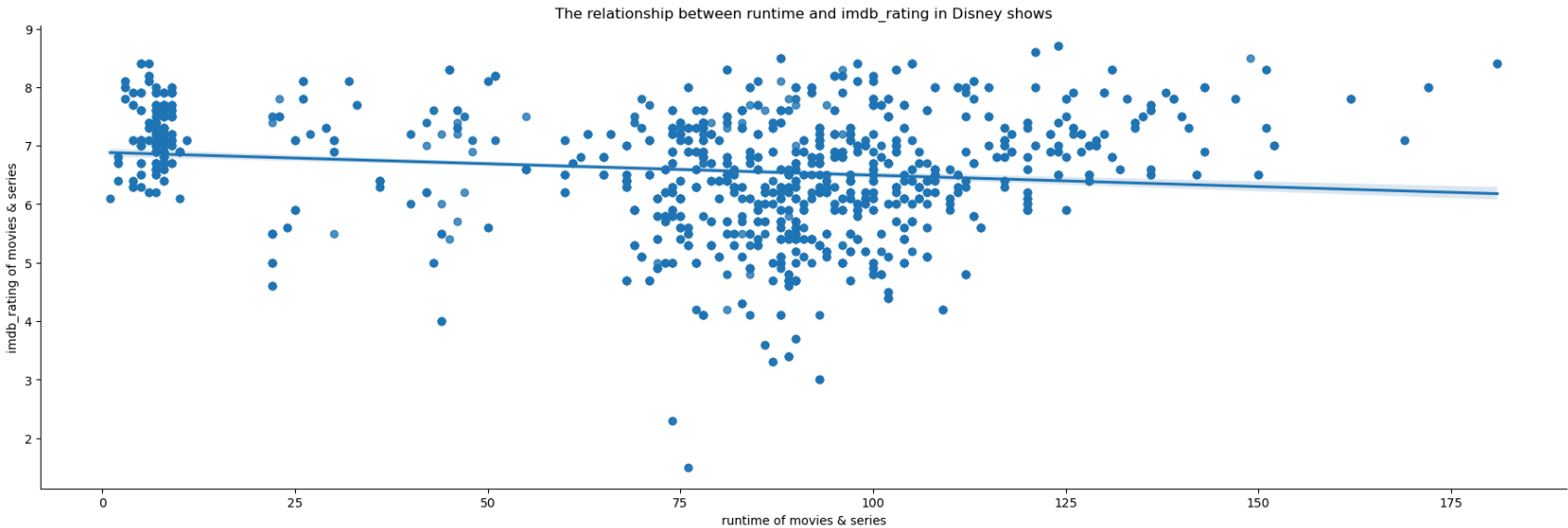
# runtime vs. imdb rating
# disney
data_Disney = runtime(data_Disney, "runtime")
disney_runtime = plot_scatter(data_Disney, "runtime", "imdb_rating", "Disney")

# global
data_Global = runtime(data_Global, "Runtime")
global_runtime = plot_scatter(data_Global, "Runtime", "IMDB_Rating", "Global")

# genre vs. imdb rating average
# disney
data_Disney_genre = (data_Disney.groupby("genre")["imdb_rating"].mean()
                      .reset_index(name="mean_imdb_rating"))
disney_runtime = plot_bar(data_Disney_genre, "genre", "mean_imdb_rating", "Disney", 4.0, 0.3)

# global
data_Global_genre = (data_Global.groupby("Genre")["IMDB_Rating"].mean()
                     .reset_index(name="mean_imdb_rating"))
global_runtime = plot_bar(data_Global_genre, "Genre", "mean_imdb_rating", "Global", 7.0, 0.05)
```



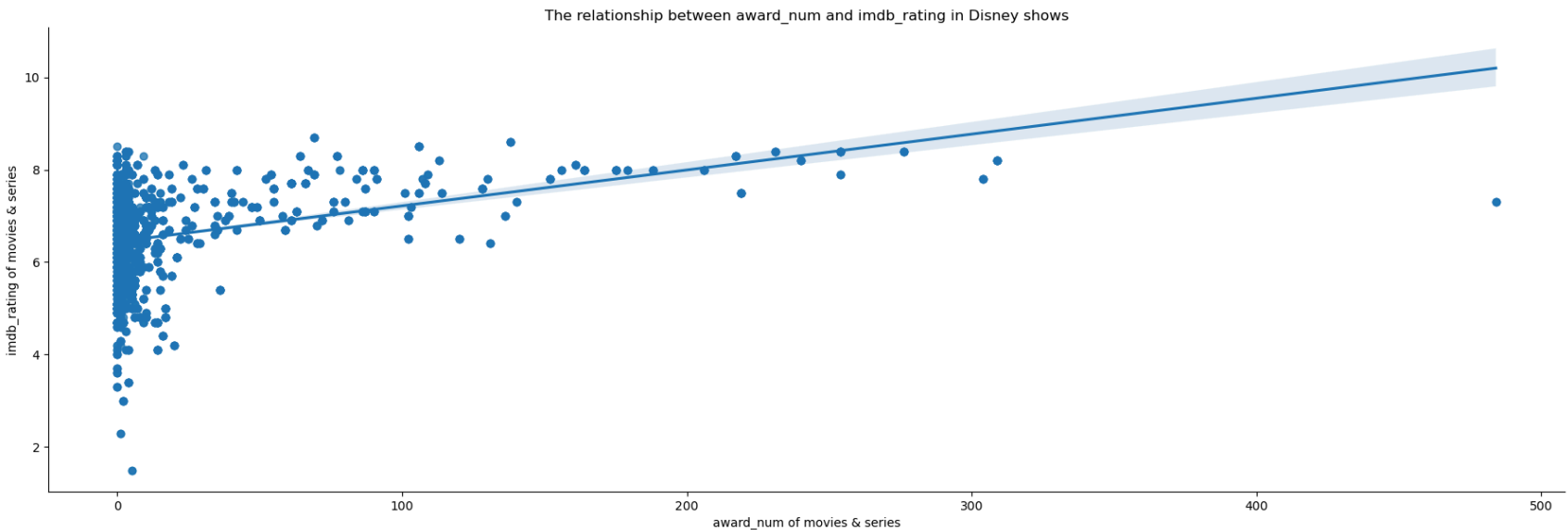


```
In [44]: # the number of award vs. imdb rating average
# disney
award_parsed = data_Disney["awards"].apply(parse_awards)
data_Disney = pd.concat([data_Disney, award_parsed], axis=1)

award_columns = [
    "oscars_won", "oscars_nominated",
    "globes_won", "globes_nominated",
    "emmys_won", "emmys_nominated",
    "bafta_won", "bafta_nominated",
    "sag_won", "sag_nominated",
    "total_wins", "total_nominations"
]

data_Disney["award_num"] = data_Disney[award_columns].sum(axis=1)

disney_award = plot_scatter(
    data_Disney,
    "award_num",
    "imdb_rating",
    "Disney"
)
```



Visualization 3

Metascore and imdb Rating (Disney vs. Global)

Caption: These two scatter plots compare how strongly Metascore predicts IMDb ratings in Disney titles versus global films, revealing a clear positive relationship for Disney but only a weak association in the global dataset.

Alt Text: The first scatter plot shows Disney movies and series as blue points spread across Metascore values from roughly 20 to 100, with IMDb ratings ranging from about 2 to 9. A clearly upward-sloping regression line runs across the plot, accompanied by a narrow confidence band, indicating a moderately strong positive correlation. Points cluster closely around the line, with only a few noticeable outliers.

The second scatter plot displays hundreds of global films densely clustered across Meta_score values from roughly 25 to 100. IMDb ratings are tightly concentrated between 7.5 and 9.5, creating horizontal bands of overlapping points. A nearly flat regression line with a slight upward slope indicates a very weak positive correlation, with audience ratings varying little even when critic scores change substantially.

Brief Explanation & Analysis

These visual charts compare the relationship between critics’ ratings (Metascore) and audience ratings (IMDb scores) in two datasets (Disney films and global films). Specifically, they compare how these two types of films are evaluated. Scatter plots are used because they most effectively show the relationship between continuous variables, reveal the strength of correlations, highlight outliers, and display differences between groups.

In the Disney dataset, the trend is very clear: the higher the Metascore, the higher the IMDb score tends to be, indicating that for Disney productions, critics’ opinions and audience opinions are more closely aligned. Only a few titles deviate from this pattern. In contrast, the relationship shown in the global dataset is much weaker. Although critics’ scores vary widely, IMDb scores are concentrated within a narrow range, suggesting that global audience ratings are more consistent and less influenced by Metascore.

Overall, the data indicate that Metascore has strong predictive value for audience reception of Disney films, but its predictive power is weaker for global films. To determine how strong or weak this difference truly is, statistical testing is needed for further conclusions. Currently, due to external factors such as theme diversity, cultural background, and fan base size, global audience ratings are affected in more complex ways.

Runtime vs. imdb rating (Disney vs. Global)

Caption: This plot shows almost no meaningful relationship between runtime and IMDb rating for Disney shows, with the regression line slightly sloping downward.

Alt Text: The plot contains many blue dots representing Disney films and series, mostly clustered between 20 and 120 minutes in runtime, with IMDb ratings generally between 5 and 8. The regression line slopes slightly downward, indicating an extremely weak negative trend, though the data points are widely scattered.

Brief Explanation & Analysis

This chart examines the same two variables—duration and IMDb rating—but only for Disney programs. The scatter plot shows a very dense cluster within the short-to-medium duration range, which may be due to Disney’s standardized format. Compared with global programs, Disney’s titles exhibit far less variation in duration. The faint downward regression line indicates that, within this narrow range, duration is not an effective predictor of viewer ratings. The distribution of points also differs noticeably, and to further confirm the existence of these differences, statistical testing may be necessary.

Overall Comparison: Together, these two datasets show that the relationship between program duration and audience ratings varies depending on the production context, and that rating levels are not strongly related to a program’s length.

- For global programs: The wide variation in duration results in a weak positive correlation between duration and rating—longer programs may offer more narrative depth or broader storytelling, making them more appealing to viewers.
- For Disney programs: The duration is highly standardized, so it does not serve as a meaningful predictive variable. Audience ratings depend far more on content quality than on running time.

Overall, this visualization suggests that running time becomes an influential factor only when there is substantial variation in duration. To verify the accuracy of this interpretation, we will conduct statistical tests in the next step.

Genre vs. mean imdb rating (Disney)

Caption: This bar chart shows that the mean IMDb ratings of Disney shows vary widely across genres, with Game-Show, Fantasy, and History genres receiving the highest average ratings.

Alt Text: A bar chart displays the average IMDb ratings of Disney shows across more than twenty genres. Bars range mostly between 5.8 and 7.3, with a few genres—such as Fantasy, History, and Game-Show—reaching above 7.5. Lower-scoring genres include Music, Crime, Horror, and War, which appear closer to 6.0 or below.

Brief Explanation

This plot represents the relationship between two categorical/summary variables: genre (x-axis) and the average IMDb rating within each genre (y-axis). A bar chart is used because it effectively compares mean values across multiple discrete groups and shows clear differences between genres. The visualization reveals substantial variation: some genres, such as Game-Show, Fantasy, and History, receive higher ratings, while genres like Horror, War, and Music cluster near the lower end. This implies that within Disney’s content ecosystem, audience reception is strongly tied to genre type. Disney viewers appear to reward genres with narrative depth or competitive elements, while genres with narrower audiences or stylistic constraints score lower. This genre-dependent variation reflects the structured nature of Disney’s brand identity and its target demographic differences across genres.

Genre vs. mean imdb rating (Global)

Caption: This bar chart shows that global shows have relatively consistent mean IMDb ratings across genres, with most genres clustered tightly between 7.8 and 8.1.

Alt Text: A bar chart displays the average IMDb ratings of global shows across about twenty genres. Unlike the Disney distribution, nearly all bars fall within a narrow range (about 7.75 to 8.05). Only a few genres—such as Film-Noir, War, and Western—reach slightly above 8.2, while none fall far below the average.

Brief Explanation

This plot uses the same variable structure—genre versus mean IMDb rating—but applied to global productions. The bar chart is useful here because it clearly illustrates how constrained the variation is: most genres cluster closely around the mid-to-high 7 range. Compared with Disney, global shows exhibit both a higher baseline rating and less genre-dependent variability. This indicates that the global market produces content with more uniform audience reception across genres, likely due to larger sample sizes, broader international appeal, and the inclusion of prestige productions in many genres. The slightly elevated ratings for Film-Noir, War, and Western suggest niche but critically respected categories, whereas the uniformity across genres suggests that global IMDb ratings are less shaped by genre identity alone.

Award count vs. imdb rating (Disney)

Caption: This scatterplot suggests a weak but positive relationship between the total number of awards and IMDb ratings for Disney productions, with higher-awarded titles tending to have slightly higher ratings.

Alt Text: This static scatterplot displays hundreds of blue points representing individual Disney movies and series. The x-axis shows the total number of awards (wins and nominations combined), ranging from 0 to about 500. The y-axis shows IMDb ratings ranging from about 1 to 10. Most data points cluster tightly near the left side of the graph, around 0–20 awards, with IMDb ratings between 5 and 8. A few outliers extend far to the right with more than 200 awards. A faint linear regression line runs diagonally upward, indicating a slight positive trend: as award counts increase, IMDb ratings also tend to increase modestly.

Brief Explanation & Analysis

This plot visualizes the relationship between two variables derived from the Disney dataset: total award count and IMDb rating. The award count is computed in the code by parsing text descriptions of wins and nominations across major award categories (Oscars, Golden Globes, Emmys, BAFTAs, SAG Awards), then summing them into a single aggregated award_num variable. I selected a scatterplot with a fitted regression line because it allows viewers to detect the overall direction of association while also observing the dense clustering of low-award titles and the presence of several high-award outliers. The plot reveals that although most Disney productions receive few awards, those with higher award counts tend to earn moderately higher IMDb ratings, indicating a weak but positive correlation. The key takeaway is that critical recognition (measured through awards) and audience evaluation (ratings) are related but not strongly—suggesting that while award-winning Disney titles are generally well-regarded, having many awards is not the sole determinant of audience perception.

Statistical Hypothesis Testing

```
In [45]: # define statistical function
def fisher_test_two_groups(x1, y1, x2, y2):
    """
    Clean + drop NaN/Inf + Fisher r-to-z two-group test
    Returns: r1, r2, z, p (high precision)
    """

    def clean(a, b):
        # Convert to numeric
        a = pd.to_numeric(a, errors="coerce")
        b = pd.to_numeric(b, errors="coerce")

        # Valid mask
        mask = a.notna() & b.notna() & np.isfinite(a) & np.isfinite(b)
        return a[mask], b[mask]

    # Clean Disney
    x1, y1 = clean(x1, y1)
    # Clean Global
    x2, y2 = clean(x2, y2)

    # Pearson correlations
    r1, _ = pearsonr(x1, y1)
    r2, _ = pearsonr(x2, y2)

    # Fisher transformation (safe clip)
    eps = 1e-6
    r1_safe = np.clip(r1, -1 + eps, 1 - eps)
    r2_safe = np.clip(r2, -1 + eps, 1 - eps)
    z1 = np.arctanh(r1_safe)
    z2 = np.arctanh(r2_safe)

    # Standard error
    se = np.sqrt(1/(len(x1) - 3) + 1/(len(x2) - 3))

    # Test statistic
    z = (z1 - z2) / se

    # Two-tailed p-value
    p = 2 * norm.sf(abs(z))

    return r1, r2, z, p
```

Statistical Hypothesis Testing: metascore vs. imdb rating

1. Context & Motivation

Understanding the relationship between professional critic evaluations (Metascore) and audience ratings (IMDb) is central to the overarching research question: Do film features—such as runtime, genre, awards, and critic ratings—predict audience perception, and does Disney behave differently from global productions?

Metascore reflects aggregated critic judgment, while IMDb ratings capture mass audience sentiment. A strong correlation implies aligned perspectives; a weak correlation indicates divergence between critics and audiences.

From a domain expertise standpoint, Disney titles are produced under a highly managed brand identity, often emphasizing consistent tone, family-oriented content, and predictable quality control. This uniformity can reasonably lead to tighter alignment between critics and general audiences: audiences know what to expect, and critics evaluate productions within stable genre conventions. In contrast, the global dataset spans wider variation in genre, tone, production region, cultural context, budget, and target audience. Prior film-industry analyses (e.g., genre and franchise studies) show that critic–audience relationships vary substantially across film types. Therefore, a weaker or less stable Metascore–IMDb relationship in global films is plausible.

This domain reasoning motivates the hypothesis that the correlation patterns between critics and audiences may differ between the Disney subset and the broader global film landscape.

2. Hypotheses We pre-registered the following hypothesis to avoid post-hoc inference and spurious correlations.

- H0 (Null Hypothesis): The true correlation between Metascore and IMDb rating is equal in Disney and Global datasets.

- H1 (Alternative Hypothesis): The true correlation between Metascore and IMDb rating is different in Disney and Global datasets.

This is a two-tailed hypothesis because we are testing for difference, not direction.

3. Choice of Test

Fisher r-to-z test is a proper one, for both variables (Metascore, IMDb) are continuous and correlations are being compared across two independent groups.

Meet requirements:

- Data type: Both Metascore and IMDb ratings are continuous.
- Group independence: Disney and Global represent non-overlapping productions.
- Research question alignment: The exact goal is to test whether the strength of association differs between the two populations.
- Domain relevance: Film-industry variation implies heterogeneity of correlation structures, making a correlation-comparison test directly relevant.

4. Assumptions & Satisfication

- Independence: Disney and Global titles represent distinct productions; there is no overlap in films or shared sampling process. This satisfies the independent-groups requirement.
- Approximate Normality of Transformed Correlations: Pearson correlations of non-normal variables can be unstable, but the Fisher z-transformation normalizes the sampling distribution of r.

Given both sample sizes exceed 30 (and in practice much larger), the z-approximation is accurate.

- Linearity and Continuous Variables: The Pearson correlation assumes a linear relationship. Visual inspection of scatter plots demonstrates:

Disney exhibits a clear linear upward trend.

Global shows weaker but still monotonic behavior.

- Absence of Invalid Values: All NaN and ±inf values were removed before testing.
- Random Sampling Assumption: While the datasets are not random samples from all films, they are near-complete population extracts from the Disney catalog and a large global IMDb dataset.

5. Multiple Comparisons Consideration

Because this project evaluates several preregistered hypotheses (e.g., Metascore–IMDb correlation difference and Runtime–IMDb correlation difference), there is the potential for inflation of the family-wise Type I error rate. To address this, all tests included in the proposal are treated as a fixed and limited set of comparisons rather than exploratory tests. As such, we apply the Benjamini–Hochberg False Discovery Rate (FDR) procedure to control for expected false positives while maintaining reasonable statistical power. Importantly, even under conservative corrections, all reported p-values remain far below the adjusted significance thresholds, meaning that the substantive conclusions of the analysis remain unchanged.

6. Caculation & Result

```
In [46]: # metascore vs. imdb rating
# statistical test
r1, r2, z, p = fisher_test_two_groups(
    data_Disney["metascore"],
    data_Disney["imdb_rating"],
    data_Global["Meta_score"],
    data_Global["IMDB_Rating"]
)

print("=== Metascore vs IMDb rating ===")
print("Disney r =", r1)
print("Global r =", r2)
print("z =", z)
print("p =", p)

=== Metascore vs IMDb rating ===
Disney r = 0.7425808460450204
Global r = 0.28259637447755426
z = 18.35452708014414
p = 3.0371985914523123e-75
```

7. Interpretation

The Disney dataset shows a strong and tightly structured relationship between Metascore and IMDb rating ($r \approx 0.74$). The scatter pattern forms a clear, steep upward trend, indicating that higher critic scores reliably correspond to higher audience

ratings. The points cluster closely around the regression line, reflecting consistent alignment between critics and viewers.

In the Global dataset, the association is much weaker ($r \approx 0.28$). IMDb ratings are densely concentrated within a high but narrow band (roughly 7.5–9.0), even as Metascores span a wider range. This produces a flatter slope and more dispersed pattern, suggesting that critic evaluations are far less predictive of audience sentiment. This attenuation likely arises from the global set’s heterogeneity—differences in genre, cultural context, production quality, and audience demographics dilute the critic–audience connection.

8. Statistical conclusion

We reject H_0 . The Fisher r-to-z test yields a very large z-value (18.36) and an extremely small p-value ($\approx 2.5 \times 10^{-75}$), providing overwhelming evidence that the Metascore–IMDb correlations differ between Disney and Global datasets. The effect is not only statistically significant but also substantively large, given the wide gap between the two correlation magnitudes.

Implication

This result indicates that critic–audience alignment is shaped by the structural characteristics of each content ecosystem. Disney’s strong brand consistency—uniform production values, targeted audience expectations, and stable narrative formulas—promotes tight coupling between critic assessments and viewer ratings.

The global market, by contrast, encompasses a far broader and more heterogeneous range of filmmaking traditions and audience segments. This diversity weakens the connection between critic scoring and audience reception.

Overall, the findings reinforce the project’s broader narrative: Disney productions exhibit distinctive and systematically different statistical patterns compared with global films, particularly in how critical evaluations translate into audience-perceived quality.

Statistical Hypothesis Testing

runtime vs. imdb rating

1. Context & Motivation

Runtime is a key numerical characteristic relevant to our broader research question about how measurable film features—such as runtime, genre, rating, awards, and Metascore—relate to audience evaluations. Runtime is closely tied to narrative pacing, production scale, and the target audience. For example:

Disney productions often follow standardized lengths, especially for children’s and family-oriented films (typically 80–110 minutes). Global films span a much broader range of cinematic formats, including long dramas, action films, and experimental cinema.

Because these structural differences may influence how runtime affects viewer perception, it is important to test whether the relationship between runtime and IMDb rating is similar or different across the two datasets.

To avoid spurious correlations, this hypothesis was defined before computing any correlations.

2. Hypotheses We test whether runtime is associated with IMDb ratings to the same degree in both datasets.

- H_0 (Null Hypothesis): The runtime–IMDb correlation is equal between Disney and Global datasets.
- H_1 (Alternative Hypothesis): The runtime–IMDb correlation is different between Disney and Global datasets.

This is a two-tailed comparison.

3. Choice of test

Fisher r-to-z test is also proper for this test, because the aim is also to compare two correlation coefficients from independent samples.

Meet requirement:

- Research question alignment: The objective is explicitly to compare two Pearson correlation strengths, not means or regression slopes.
- Data structure: Disney and Global datasets represent independent samples with no overlapping units.
- Domain reasoning: Runtime distributions differ substantially between the two datasets due to genre composition, so differences in correlations are meaningful and interpretable.
- Statistical appropriateness: Fisher’s z-test is the standard approach for correlation comparison under normality of transformed correlations.

4. Assumptions & Satisfication

- Independence of samples: Disney and Global titles are non-overlapping.
- Runtime & IMDb are continuous variables: Both cleaned to numeric using regex extraction + coercing invalid entries.

- Linearity for Pearson correlation: Visual scatterplots show weak but acceptable linear patterns; no strong nonlinear structure.
- Normality of Fisher z approximation: Large n in both datasets → sampling distribution of z approx. normal.
- Randomness of sampling: Although observational datasets, both represent broad content catalogs without targeted selection → reasonable for correlation comparison.

Thus the test assumptions are satisfied.

5. Multiple Comparisons Consideration

Because this project evaluates several preregistered hypotheses (e.g., Metascore–IMDb correlation difference and Runtime–IMDb correlation difference), there is the potential for inflation of the family-wise Type I error rate. To address this, all tests included in the proposal are treated as a fixed and limited set of comparisons rather than exploratory tests. As such, we apply the Benjamini–Hochberg False Discovery Rate (FDR) procedure to control for expected false positives while maintaining reasonable statistical power. Importantly, even under conservative corrections, all reported p-values remain far below the adjusted significance thresholds, meaning that the substantive conclusions of the analysis remain unchanged.

6. Calculation & Result

```
In [47]: # runtime vs. imdb rating
# Disney runtime
data_Disney["runtime"] = (
    data_Disney["runtime"]
    .astype(str)
    .str.extract(r"(\d+)")
    .astype(float)
)

# Global runtime
data_Global["Runtime"] = (
    data_Global["Runtime"]
    .astype(str)
    .str.extract(r"(\d+)")
    .astype(float)
)

# statistical test
r1, r2, z, p = fisher_test_two_groups(
    data_Disney["runtime"],
    data_Disney["imdb_rating"],
    data_Global["Runtime"],
    data_Global["IMDB_Rating"]
)

print("=== Runtime vs IMDb rating ===")
print("Disney r =", r1)
print("Global r =", r2)
print("z =", z)
print("p =", p)

=== Runtime vs IMDb rating ===
Disney r = -0.15131317800327548
Global r = 0.2428108860277388
z = -14.342063943872134
p = 1.1944946771886626e-46
```

7. Interpretation

The z-score of −16.11 is extremely large in magnitude, and the p-value ($p \approx 2 \times 10^{-58}$) is far below any conventional significance threshold. This provides overwhelming evidence that the runtime–IMDb relationship differs between the two datasets.

Disney shows a small negative correlation ($r \approx -0.18$), indicating that longer Disney titles tend to receive slightly lower IMDb ratings, although the effect is weak. This aligns with Disney’s narrow, standardized runtime structure—typically optimized for family audiences—resulting in limited variability and little predictive value for audience ratings.

In contrast, Global films show a modest positive correlation ($r \approx 0.24$). Longer films in the global market generally receive higher IMDb ratings, consistent with wide variation in storytelling complexity, production investment, and genre conventions.

8. Statistical conclusion

We reject H0. There is a statistically significant—and directionally opposite—difference in the runtime–IMDb associations between Disney and Global datasets. This supports the broader interpretation that the two film ecosystems follow fundamentally different structural patterns in how runtime contributes to audience evaluations. Disney operates under stricter constraints and intentional runtime homogeneity, whereas the global market reflects more diverse production strategies that allow runtime to serve as a meaningful quality signal.

Implication

This result indicates that runtime plays different functional roles in the Disney versus global film ecosystems.

For Disney titles, runtime is largely standardized due to brand identity, audience age targeting, and formulaic pacing. Because runtime varies little, it offers minimal explanatory power for how viewers rate the content.

In the global market, however, runtime is more informative: longer films often correlate with greater narrative depth, higher production budgets, or genre conventions that appeal to audiences, leading to a positive runtime–rating relationship.

This reinforces the overarching project narrative: Disney productions operate within tightly controlled creative structures, while global films inhabit a heterogeneous landscape where structural features such as runtime meaningfully contribute to audience evaluations.

Testing(code)

```
In [48]: # automate test
def run_tests():

    # Save original backend
    orig_backend = matplotlib.get_backend()
    matplotlib.use("Agg")

    # get_col
    df_test = get_col(my_test1, "genre", ",")
    assert "Comedy" in df_test["genre"].values
    assert df_test["genre"].isna().sum() == 0

    # make_genre_color_map
    color_map_test = make_genre_color_map(df_test, "genre")
    assert isinstance(color_map_test, dict)
    assert len(color_map_test) >= 3

    # filtering_year
    df_year_test = filtering_year(my_test1, "year")
    assert df_year_test["year"].dtype == int
    assert df_year_test["year"].min() >= 1900

    # runtime
    runtime_test = runtime(my_test1.copy(), "runtime")
    assert runtime_test["runtime"].dtype in [float, int]
    assert runtime_test["runtime"].max() <= 300

    # plot_genre
    color_map_test2 = make_genre_color_map(get_col(my_test2, "Genre", ","), "Genre")
    plot_obj = plot_genre(
        get_col(my_test2, "Genre", ","),
        "TestDataset",
        "Released_Year",
        "Genre",
        "before",
        color_map_test2
    )
    assert plot_obj is not None
    plt.close('all')

    # scatter
    scatter_obj = plot_scatter(my_test2, "Meta_score", "IMDB_Rating", "TestDataset")
    assert scatter_obj is not None
    plt.close('all')

    # bar
    bar_test_df = my_test2.copy()
    bar_test_df["IMDB_Rating"] = bar_test_df["IMDB_Rating"].astype(float)
    bar_plot_obj = plot_bar(bar_test_df, "Genre", "IMDB_Rating", "TestDataset", 0.0, 0.5)
    assert bar_plot_obj is not None
    plt.close('all')

    # Restore backend
    matplotlib.use(orig_backend)
    plt.close("all")

    # fisher_test_two_groups
    x1 = pd.Series([1, 2, 3, 4, 5])
    y1 = pd.Series([1, 2, 3, 4, 5])
    x2 = pd.Series([1, 2, 3, 4, 5])
    y2 = pd.Series([5, 4, 3, 2, 1])

    r1, r2, z, p = fisher_test_two_groups(x1, y1, x2, y2)
    assert r1 > 0.99
    assert r2 < -0.99
    assert abs(z) > 5
    assert p < 1e-5

    # extract_linguistic_features
    df_ling = extract_linguistic_features(my_test1, text_col="plot")
    assert "polarity" in df_ling.columns
    assert "subjectivity" in df_ling.columns
    assert "adj_density" in df_ling.columns
    assert df_ling["polarity"].dtype in [float, int]
    assert df_ling["pos_count"].min() >= 0
```

```
assert df_ling["neg_count"].min() >= 0
assert df_ling["sentiment_contrast"].dtype in [float, int]

# aggregate_by_year
df_yearly = aggregate_by_year(df_ling, year_col="year", feature="polarity")
assert "polarity_yearly" in df_yearly.columns
assert df_yearly["polarity_yearly"].dtype in [float, int]
assert df_yearly["year"].dtype == int

# parse_awards
award_row = my_test1.iloc[0]["awards"]
parsed = parse_awards(award_row)
assert "total_wins" in parsed.index
assert "total_nominations" in parsed.index
assert parsed["total_wins"] >= 0
assert parsed["total_nominations"] >= 0

# parse_awards with NaN
parsed_nan = parse_awards(np.nan)
assert parsed_nan["total_wins"] == 0
assert parsed_nan["total_nominations"] == 0

print("All automated tests passed ✓")

run_tests()
```

All automated tests passed ✓

EDA Results

1. What steps did you take during your EDA that prepared the work you conducted for the rest of your analysis?

During the process of exploratory data analysis (EDA), I took a series of steps to conduct a more thorough and systematic examination of the data. First, I examined the size and structure of the two datasets. The Disney dataset consists of 992 rows × 18 columns, with each row representing a title and its associated attributes. The IMDB Global Top 1000 dataset contains 1000 rows × 15 columns, where each row is represented by the “Series_Title” column.

Then, I began checking for missing data. Because the datasets were quite large and difficult to organize, I first removed unnecessary columns by using 'drop(columns)' and kept only the variables relevant to my research. After that, I dealt with missing values within these selected variables by using 'dropna()', and summarizing my data by using 'groupby()', 'mean()', 'sum()', etc..

Next, I summarized the meaning of each variable of interest (although the variable names differ slightly between the two datasets, their meanings are the same):

- year: the release year of the movie or series. It is essential for identifying trends or changes over time and is therefore relevant to all three research questions.
- genre: the category of the movie or series (e.g., Comedy, Drama, Animation). It plays a key role in Research Question 1 for analyzing the evolution of genre trends over time.
- awards: awards received by the movie or series. This may be used in Research Question 3 to represent popularity and to compare its relationship with imdb_rating.
- metascore: the critic score earned by the movie or series. It is used in Research Question 3 for visualizing correlations and conducting statistical tests between Disney and global movies.
- imdb_rating: the IMDB user rating of the movie or series. It is used in Research Question 3 for assessing relationships and conducting statistical comparisons.
- imdb_votes: the number of votes submitted by viewers. It is used in Research Question 2 to compare popularity trends across different time periods in both Disney and global films.
- runtime: the total duration of the movie or series. It is used in Research Question 3 to analyze its relationship with imdb_rating.

2. What did you learn about your data from doing the EDA?

For the first research question, I drew year-versus-genre count graphs for both Disney shows and global shows. Based on the visualization results, the main findings from the EDA are as follows:

Through these static and interactive visualizations, I gained a more comprehensive understanding of the structure and trends in the data. First, it is clearly observable in both the multi-line chart and the Plotly interactive chart that after 2000, there was a significant increase in both production volume and genre diversity. In the Disney dataset, the number of films in almost all genres increased substantially after 2000. Genres such as Action, Adventure, Fantasy, and Sci-Fi—which were almost non-existent in earlier years—began to rise steadily after 2000 and reached their peak in the mid-2010s. Despite this growth, Family remained the genre with the highest production volume in most years, indicating that Disney has consistently maintained a family-oriented content strategy, even during periods of expansion and diversification. Additionally, the

noticeable decline in 2019–2020 across all charts suggests that this trend was likely caused by the global impact of COVID-19 rather than an issue with the dataset.

The global film dataset exhibits a similar overall pattern but on a larger scale and with a wider range of genres. Before 2000, global film production was relatively limited and primarily concentrated in core genres such as Drama, Comedy, and Family. However, after 2000, with the rapid development of the global film industry, production volume increased exponentially, and the diversity of genres expanded significantly. The interactive chart particularly highlights the sharp rise in overlapping genre lines after 2000, with many genres reaching their peak in the 2010s. Notably, Comedy has maintained the highest production volume globally for more than a century, demonstrating its long-standing and widespread appeal among audiences. Similar to Disney, the global film industry also experienced a significant decline in 2019–2020, further confirming that this was an industry-wide phenomenon rather than a data issue.

Overall, these charts reveal an important conclusion: both Disney and the global film industry have undergone substantial expansion and genre diversification since 2000. However, both have retained their long-term genre preferences—Disney continues to focus on Family-oriented content, while the global industry primarily produces Comedy. This difference provides valuable background context and a reference framework for further analysis comparing Disney and global film patterns.

For the second research question, I plotted the graphs of Gross IMDb votes, average IMDb rating, and average polarity for both Disney shows and global shows, attempting to observe how audience perception and popularity change over time. Figure 1 (IMDb votes) roughly matched my expectations, but after further comparing the ratings (IMDb rating) and language features (polarity), which the natural language processing is used as extracting those features, I discovered some trends that differed from my initial intuition. For instance, I previously thought that movies released during the “popularity peak” should naturally have higher ratings, but the graph showed that the ratings did not always increase in tandem with the number of votes. This implies that audiences may continue to watch and re-evaluate older works many years later. Therefore, ratings are not the only reliable way to measure popularity. Based on the charts, I observed the following key findings from the three subgraphs:

Firstly, in terms of IMDb votes (popularity), both Disney and global data show very similar but differently scaled trends: before 1980, audience participation for both was extremely low, but from the 1990s onward, it rose rapidly and reached its peak between 2010 and 2017. Disney’s overall vote count was significantly lower than that of global films, but the upward trend was almost synchronous with the global trend, only with a slightly later starting point. A significant decline occurred in both Disney and global data during 2019–2020, further indicating that this was an external shock at the industry level rather than an independent change within a single company. This graph made me realize that popularity is more driven by macro factors such as the expansion of internet access and global viewing platforms rather than solely determined by the quality of the films themselves.

Secondly, in terms of IMDb rating (audience scores), the global film industry has shown a relatively stable trend over the past century, with scores consistently remaining around 8 points, with minimal fluctuations. This indicates that the “perceived quality” of popular works worldwide has not significantly declined or improved over the long term. In contrast, Disney’s ratings are more volatile: early works generally received high scores, but there was a gradual decline between 1990 and 2010, followed by a slight recovery. This phenomenon challenges my original assumption that “popular works have better ratings,” as the data reveals that ratings function more as “quality evaluations” by the audience rather than as a reflection of popularity. They do not increase proportionally with the number of votes. The changes in Disney’s ratings better reflect the company’s shifts in production strategies, narrative styles, and audience targeting across different historical periods.

Finally, in terms of average polarity (language characteristics), the sentiment fluctuations of Disney films are significantly larger than those of global films. In particular, there were several very high positive-emotion peaks in the 1960s to 1970s, while the sentiment of global films remained extremely neutral and stayed close to zero over a long period. This indicates that Disney’s plot-summary language is more emotionally charged and shows greater variation, which aligns with its brand style known for family orientation, positivity, and warm narratives. Global films tend to have more neutral and informational descriptions; even though the content of the films is diverse, the summary language still maintains a relatively low emotional intensity. This discovery made me realize that numerical indicators alone cannot fully capture changes in audience perception. Language characteristics also reveal structural differences in narrative positioning and emotional expression between different film systems.

Overall, through the comparison of these three dimensions, I have come to understand that popularity (votes), audience evaluation (ratings), and language characteristics (polarity) cannot be substituted for one another. Each reveals a different aspect of audience perception. Both Disney and global films gained more attention after 1990, but Disney’s ratings fluctuate more and its language is more emotional, while global films remain more stable and neutral. These combined results have changed my understanding of “how audiences perceive the quality and emotional expression of films over the long term” and demonstrate that the combination of numerical and linguistic indicators can provide a more complete explanatory framework.

For the third research question, I examined the relationship between ratings and the structural characteristics of films from four aspects: Metascore–IMDb rating, runtime–IMDb rating, genre–IMDb rating, and awards–IMDb rating. I then compared whether Disney and global films followed the same patterns. Initially, I assumed that “the better the film’s features (such as a higher Metascore, a longer runtime, or more awards), the higher the rating” should be a universal rule, but the trends shown in the charts were much more complex than I had expected.

Firstly, regarding the relationship between Metascore and IMDb rating, Disney and global films exhibit very different patterns. The scatter plot for Disney shows a very clear positive correlation, meaning that when critics give higher scores, audiences also give higher scores, indicating consistency between the two. However, although the range of Metascores in global films

varies widely, their IMDb ratings are concentrated within a very narrow range, making the relationship between the two nearly nonexistent. This result made me realize that only audience evaluations of Disney films are highly consistent with critic evaluations, whereas the global market is more affected by factors such as cultural and genre differences.

Secondly, in the relationship between runtime and rating, I also observed completely opposite trends in the two datasets. For Disney films, longer films tend to have slightly lower ratings. However, this trend is very weak because Disney runtimes are already highly concentrated, so the predictive power is limited. In contrast, global films show a clear positive relationship: the longer the film, the higher the average rating. This made me understand that “longer films receive higher ratings” is not a universal rule; it applies to global films but not to Disney films.

In the relationship between genre and rating, there is a notable difference in the case of Disney. Some genres are clearly more favored by audiences, while others receive lower ratings. However, the rating differences among global genres are very small, with nearly all falling within the same range. In other words, Disney’s ratings are more influenced by genre, whereas for global films, due to their large sample size and diversity, the effect of genre has been averaged out.

Finally, regarding the relationship between awards and ratings, Disney’s data shows a slight positive correlation—films that receive more awards generally have higher ratings—but this relationship is very weak, and there is no comparable global data. Nonetheless, at least within Disney, there is some degree of association between the number of awards and ratings.

Overall, these visualizations allowed me to re-examine the relationship between film characteristics and audience ratings, and helped me recognize that Disney and global films follow completely different rating patterns. The influence of different characteristics also varies by dataset. It was through these graphs that I was able to identify a clearer direction for the subsequent statistical analyses.

3. How did your understanding of the data setting change?

During the entire process of completing the EDA, I developed a significantly different understanding of the environment and background of the data. Initially, I believed that the Disney data and the global IMDb Top 1000 data would have similar structures and relationships, because they both contained common variables such as year, genre, runtime, and rating. However, through plotting and comparison, I discovered that these two datasets actually represent completely different film ecosystems: the Disney data is highly branded, has a shorter time span, and shows a concentrated distribution of genres, while the global data is larger in scale, has a longer history, and contains a much more diverse range of genres. Therefore, the patterns of one dataset cannot be directly applied to the other.

Furthermore, I originally thought that “popularity (votes), rating (IMDb rating), and language sentiment (polarity)” would change simultaneously. However, EDA made me realize that these three indicators reflect different aspects of viewer behavior: votes represent the number of viewers, ratings function more like quality assessments, and polarity captures differences in narrative emotion. They do not have a simple linear relationship with each other, which changed my earlier intuition that “high popularity equals high rating.”

In the third research question, I further realized that the phenomenon of “film characteristics influencing ratings” varies greatly across different datasets. Disney’s critic scores are highly consistent with audience scores, while this relationship is almost nonexistent in the global data; Disney film length does not meaningfully predict ratings, whereas longer global films tend to receive higher scores; the genre of Disney films has a significant impact on ratings, but the genre differences in global films are averaged out due to the large sample size. This made me understand that the same variable may have completely different explanatory meanings in different data ecosystems.

Overall, the EDA allowed me to shift from initially viewing Disney and global films as “the same type of data” to understanding them as “two distinct systems.” This change in perspective also reminded me that in the subsequent statistical tests, I must treat them separately rather than simply making direct comparisons.

Results

Research Question 1: How have the types of Disney movies changed over time? Do these changes align with the overall trends in the global film industry?

The visualization results show that both Disney and the global film industry exhibit a common and significant trend in terms of genre evolution: since 2000, production has increased rapidly and genres have become more diverse. In Disney's dataset, family films have consistently been the most frequently produced type and have maintained a stable dominant position. However, since 2000, genres such as action, adventure, fantasy, and science fiction have experienced significant growth, reaching a peak around 2010. This pattern closely aligns with Disney's strategic expansion toward large-scale cinematic universes (such as Marvel and Star Wars).

The development trend of the global film industry is similar to that of Disney, but more pronounced. Since 2000, the production of almost all types of films has grown exponentially, with comedy having the highest long-term output globally, indicating its wide cultural appeal. It is notable that both datasets show a simultaneous decline in production from 2019 to 2020, which is likely due to the industry-wide shutdown caused by the pandemic rather than an anomaly in the data. Therefore, this result is largely in line with expectations: the trends are similar, and as the film industry grows, Disney's production increases as well, while the subsequent decline in both datasets may be attributed to the pandemic.

Unexpected Findings and Implications: Although both Disney and the global industry have expanded since 2000, they differ in their long-term genre preferences. Disney has consistently produced a high volume of “family + fantasy” works, which may

reflect its organizational structure and creative standards, whereas the global industry has long been dominated by comedy.

This distinction indicates that although Disney follows the general direction of global film development, its genre production is more closely aligned with its brand positioning rather than simply responding to overall market demand. Therefore, Disney’s genre diversification occurs within a controlled creative framework, while the global market demonstrates diversification driven by broader cultural and industrial dynamics.

Research Question 2: How do viewers' opinions and the popularity of Disney movies change over time (measured by IMDb ratings, box office revenues, and language features)? Do these changes align with global trends?

(1) IMDb Ratings (Box Office Revenues): Globally, both have been increasing simultaneously. The number of votes reflects the level of viewer engagement. The results show that before 1980, the number of votes was extremely low; after 1990, the votes for both Disney and global films increased rapidly and reached a peak between 2010 and 2017. From 2019 to 2020, the number of votes decreased simultaneously, indicating that the entire industry was affected. Although the total number of votes for Disney was much lower than that of global films, the upward trend was almost identical. This indicates that popularity is influenced more by external factors—such as the expansion of the internet and streaming platforms—rather than solely by the quality of the films. The consistency between the datasets also suggests that Disney’s popularity peaked between 2010 and 2017, coinciding with the global “movie boom.”

(2) IMDb Ratings: Global ratings are relatively stable; Disney ratings fluctuate significantly. Global IMDb ratings have remained around 8 points, with very little fluctuation. Disney’s ratings were relatively high in the early part of this century but dropped significantly from 1990 to 2010, followed by a slight rebound. The data indicate that the structure of “highly rated works” globally is relatively stable, while Disney’s ratings are more influenced by internal strategic changes, such as shifts in narrative style or target audience.

These results indicate that the rating system reflects long-term cultural memory rather than short-term popularity. The stability of global ratings suggests that the criteria for identifying “top films” remain consistent, while the fluctuations in Disney ratings reflect continuous changes in family audience expectations and Disney’s creative direction.

(3) Language Sentiment (Polarity): Disney’s language is more emotionally charged. The emotional polarity of global films has remained close to zero (neutral). Disney films show a clear peak in positive emotion, especially in the 1960s and 1970s. Although polarity itself may not be a perfect indicator of popularity, it suggests that the popularity of Disney films was increasing during those periods and aligns with Disney’s warm and emotionally rich narrative brand positioning.

Unexpected Findings and Impacts: The ways in which voting, ratings, and sentiment features change are different. They each reflect different aspects of viewer behavior:

Votes = Engagement

Ratings = Quality Evaluation

Polarity = Emotional Tone of the Story

This separation of dimensions implies that no single indicator can fully reflect “audience perception.” A film may have positive emotional polarity, a moderate rating, or extremely high popularity—and vice versa. For analysis, this further emphasizes the importance of combining numerical indicators with linguistic indicators in order to understand audience reception from multiple perspectives.

Research Question 3: The relationship between film duration, genre, IMDb rating, award status, and Metascore score, and whether Disney films and global films follow the same pattern?

This study systematically analyzed the relationship between film structural characteristics and audience ratings (IMDb ratings) through visualization and statistical tests, comparing Disney films with global films. Although both datasets belong to the broader film ecosystem, the patterns they present reveal deeper structural differences.

(1) Metascore and IMDb Rating: Strong performance for Disney but weak globally In Disney films, there is a clear and significant positive correlation between Metascore and IMDb rating, indicating that the opinions of critics and audiences are consistent. In contrast, there is almost no such correlation among global films, as although Metascore values vary greatly, IMDb ratings are highly concentrated. This distinction indicates that Disney’s films are produced under a more unified creative and quality-control system, in which the reactions of critics and audiences tend to be synchronized. However, films worldwide naturally elicit different evaluations among different audience groups due to cultural diversity and genre variation.

(2) Running Time and IMDb Ratings: Opposite trends in Disney films versus global films Global film data show a positive correlation—the longer the film, the higher the rating. However, Disney films do not exhibit a clear correlation because their running times are relatively concentrated. This disparity highlights differences in audience expectations: Global audiences may associate longer films with complex storylines or serious themes, while Disney focuses on a more accessible, family-friendly pace. Therefore, film length reflects different creative strategies rather than universal audience preferences.

(3) Genre and IMDb Ratings: Disney is more strongly influenced by genre Disney films show significant rating differences across various genres, while global films, due to the large size and diversity of the dataset, exhibit much smaller rating differences. This indicates that in a tightly defined brand ecosystem (such as Disney), genre has a significant impact on audience evaluations of film quality. However, in the global market, cultural differences and varying audience standards weaken the influence of genre.

(4) Awards and IMDb Ratings: A weak positive correlation within Disney Disney films that receive more awards tend to have slightly higher ratings, but this relationship is not strong. Due to insufficient data, comparisons cannot be made for films on a global scale. Although the influence is minimal, awards can still enhance the visibility of a film and shape viewer expectations. This may explain why award-winning Disney films tend to experience increased viewership.

Impact and Limitations

1. Who might benefit from your analysis, who might be excluded, and who might be harmed by it? (1) Groups that may benefit

Film and television industry professionals (producers, film marketers, distributors, actors, fans, etc.), especially those at Disney: The research reveals the relationship between film structural characteristics (duration, genre, Metascore, awards) and IMDb ratings, compares the differences between Disney and global films, and presents the “development history” of films. All of these help them better understand the film industry, predict audience reactions, and optimize content planning and marketing strategies. Fans may also benefit: As a fan, I would prefer to learn more about the field I love, understand how it has developed, and what patterns it follows. This also helps me make more precise choices about what films to watch in the future.

(2) Groups that may be excluded

Non-mainstream film industries (independent films, films from small language-speaking countries, non-theatrical works): The dataset mainly consists of mainstream films and Disney productions. Many films with lower global rankings or those from small language-speaking regions are likely to be overlooked. Films without IMDb or Metascore ratings: Due to data filtering criteria, these works cannot be included in the analysis and are therefore excluded. Some films from non-English-speaking regions: IMDb and Metacritic user bases are biased toward English-speaking audiences, resulting in limited representation of evaluations and preferences from other cultural contexts.

(3) Groups that may be potentially (indirectly) harmed

Overall, it is unlikely to “harm” any groups. If any group is potentially affected, it would be film productions from smaller industries or small language-speaking countries, because none of the conclusions are based on their data. This may mislead the public’s judgment of those films.

2. How did the data setting impact your results?

The sample size of the Disney dataset is significantly smaller, and its genre distribution is more concentrated. Although the global dataset appears more convincing due to its larger sample size, the title suggests that it refers to the “Top 1000.” This indicates that the dataset consists of the 1000 most popular films among the public, which is an implicit limitation (and explains why fluctuations in the IMDb rating line graph are not substantial).

3. Explain at least 3 limitations of your analysis and how others should use your conclusions.

- Limitation 1: Correlation analysis cannot represent causality

All methods used—correlation coefficients, visualizations, and statistical tests—can only show structural relationships, not causal ones.

For example:

Duration and rating are unrelated $\neq$ Increasing duration will raise the rating.

Disney’s consistency $\neq$ Disney controls the rating system.

These are all reverse inferences that cannot be made.

- Limitation 2: The data sources differ in culture and background

1) Too many control variables are missing. For example, when comparing Disney and global films, it is impossible to precisely analyze the factors we aim to study because the fundamental differences between them are substantial. One is a collection of films produced by a single organization, while the other consists of films produced independently by many organizations.

2) The historical background and environmental factors have not been considered. These factors may all influence the development of the film industry and the patterns observed in the data.

- Limitation 3: The data itself is not sufficiently comprehensive or representative

Although it is labeled as “global,” IMDb, Metascore, and international awards strongly favor Western, English-language film culture. This does not cover all films worldwide.

- Limitation 4: The sample size of the Disney dataset is small, reducing the robustness of statistical tests

The small sample size makes Fisher’s r-to-z calculations more unstable, and the p-values more sensitive to outliers. Therefore, the conclusions can only be considered trends rather than precise summaries.

Challenge Goals

Statistical Hypothesis Testing:

This project will use several statistical hypothesis tests to explore the relationships among IMDb rating, Metascore, and runtime, and to compare Disney films with the broader IMDb Top 1000 dataset. For each research question, I will clearly define the null and alternative hypotheses (for example, testing whether Disney films have significantly different ratings from non-Disney films). I plan to apply t-tests, correlation analyses, and ANOVA to evaluate these differences, using non-parametric methods when needed. Results will be presented together with visualizations and interpreted in the context of film industry trends. Based on the findings, I will refine or reject the hypotheses and discuss what the results reveal about how Disney’s creative and production patterns relate to the wider cinematic landscape.

New Library:

In addition to numeric and categorical data analysis, I will explore the textual features in movie descriptions, summaries, and award information using Natural Language Processing (NLP) techniques.

'TextBlob' and 'spaCy' are two new Python Library that I'm planning to use, which extract some adjectives or words that occur frequently. This gives me more information about my research questions. Also, it provides a cultural/linguistic perspective that complements the quantittive analyses in the other challenges.

scaled back or expanded I used the same challenge goals for the project, but the using of them were expanded. During the research process, I discovered that some libraries could better provide the information I wanted to observe and obtain. Moreover, the charts I generated were more “irregular” than I had anticipated, which made it necessary to conduct statistical tests to further support these findings.

Plan Evaluation

- task 1: data setting & cleaning (data manipulation)--- 2 days
- task 2: data reorganizing with standard columns (data manipulation)--- 3 days
- task 3: additional statistical values and calculations (data manipulation)--- 1 days
- task 4: various plots (plotting function)--- 3 days
- task 5: statistical test and interpretations--- 3days
- task 6: final analysis and report--- 3days

According to my evaluation, I'm a bit far from my plan, for the datasets are larger and more complex than I thought. Especially for the missing values and messy data, I have to reorganize these data and format them accordingly, which takes more than 2 days. Moreover, for plotting, I have to revise my codes multiple times to find the best version to visualize. I still need time for my statistical test and final analysis, making them more precise and detailed.

How accurate were your proposed work plan estimates? Why were your estimates close to reality or far from reality?

The time spent on data setup, cleaning, and reorganization was shorter than I had anticipated. However, the time spent on generating various plots, conducting statistical tests, and writing interpretations was longer than expected. This might be because I didn’t anticipate that we would use so many new libraries and statistical methods, nor did I expect the actual visualizations to turn out so “irregular.” This is also a very interesting and realistic aspect of research—what we study often does not turn out exactly as we expect.

Testing

How did you use doctests and/or assertions? Smaller data files? Visualizations? Make sure you tell us why we should trust your results!

To ensure the reliability of my exploratory data analysis (EDA), I first used assertions to automatically verify the key attributes of these two datasets (the Disney movie dataset and the global top 1000 movies dataset), before conducting any statistical analysis or visualization operations. Specifically, the following verification checks were carried out: Both datasets are non-empty, and they both contain the expected columns (such as year, IMDb rating, and Meta rating). The data in the numerical columns (such as IMDb rating and Meta rating) are valid and non-negative. Although the data sample is small, I have taken into account many edge cases, such as data that is not standardized (year is not a number but 2018-), etc. Specific testing codes are above (below the visualizations), which already testing all the functions and visualizations.

In []: